Macroscopic Quantities from Microscopic Dynamics in Equilibrium Statistical Mechanics

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Chapter 1

Introduction

Statistical mechanics serves as the fundamental bridge between the microscopic dynamics of many-body systems and their macroscopic thermodynamic behavior[1]. Traditionally, this connection is formalized through ensemble theory, which relies heavily on the ergodic hypothesis and the equal a priori probability postulate to evaluate macroscopic observables without solving the intractable $6N$-dimensional Hamilton equations. While the equivalence of statistical ensembles such as the microcanonical and canonical is a well-established textbook result in the thermodynamic limit[2], directly connecting these statistical constructs to the underlying phase space geometry and Hamiltonian dynamics remains a rich area of theoretical and computational physics.

In this report, we investigate the dynamical and geometric foundations of thermodynamics. Specifically, we explore how thermodynamic quantities, such as entropy and temperature, can be rigorously defined and calculated purely from the dynamical evolution of an isolated system. Building upon the mechanical interpretation of entropy via adiabatic invariants and the Helmholtz theorem, we delve into the Rugh temperature formalism. This geometric approach allows temperature to be expressed as a phase-space divergence evaluated on the microcanonical energy hypersurface, offering a robust method to compute thermal observables as time-averages along dynamical trajectories. Another advantage of our approach is that we can consider the thermodynamics of finite systems, hence we are able to determine the corrections for thermodynamic quantities due to the finite N. These corrections do not arise in usual textbook thermodynamics since standard thermodynamics is formulated explicitly within the thermodynamic limit $N, V \to \infty$.

To validate these theoretical frameworks, we perform extensive numerical simulations using symplectic integrators. These integration schemes strictly preserve the underlying symplectic geometry and conserved quantities of the

[1] For a brief review of the history of thermodynamics, see [1].
[2] For instance, see [2].

Hamiltonian flow, circumventing the artificial energy drifts common in standard numerical methods[3]. We apply this machinery to the mean-field ϕ^4 theory, analyzing its caloric curves, phase transitions, and symmetry-breaking mechanisms. Furthermore, we extend our dynamical approach to constrained systems possessing multiple conserved quantities by employing the Dirac-Bergmann formalism and the Federer-Laurance derivation formula.

Finally, we challenge the standard assumptions of classical mechanics by investigating systems with a modified, bounded kinetic energy of the form $K_p = 1 - \cos p$. Such kinetic energies arise in tight-binding models of electrons, see [4]. Utilizing the Hubbard-Stratonovich transformation and saddle-point approximations, we reveal exotic thermodynamic phenomena in both the modified mean-field ϕ^4 and ϕ^6 models and the one-dimensional XY model[4]. Most notably, the bounded nature of the phase space leads to an entropy that decreases at high energies, resulting in the emergence of negative absolute temperatures and anti-aligned spatial correlations.

The report is organized as follows: Chapter 2 provides a brief review of the microcanonical and canonical ensembles. Chapter 3 develops the dynamical approach to thermodynamics and analyzes the mean-field ϕ^4 model under both unconstrained and constrained dynamics. Chapter 4 examines the statistical mechanics of systems with modified kinetic energy. Finally, Chapter 5 presents our concluding remarks. Also, in appendices there are various codes that used in the text and a brief introduction to the ergodic theory of dynamical systems.

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[3] For a discussion of this issue, see the first chapter of [3].

[4] For the importance and comprehensive discussion of the original XY model from an integrability point of view, consult the first chapter of [5].

Chapter 2

Basics of Microcanonical and Canonical Ensemble

The aim of statistical mechanics is to understand equilibrium and non-equilibrium states of many-body systems. Such systems may consist of N particles contained in a volume V where typically

$$N \sim 10^{23} \text{ molecules} \qquad V \sim 10^{23} \text{ molecular volumes} \sim 10 \text{ liters for a gas}$$

Thus, practically we can consider the limit $N, V \to \infty$ with N/V fixed, *i.e.* the so called *thermodynamic limit*.

A classical system made of N particles can be described by means of the $6N$-dimensional phase space

$$(q_1, \ldots, q_{3N}; p_1, \ldots, p_{3N}) := (q, p)$$

The time evolution of the system is determined by the Hamiltonian function $H(q,p)$ by means of the Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \tag{2.1}$$

For $N \sim 10^{23}$, there is no way to solve this system of ODE's except for some special integrable cases (e.g. free particles, coupled harmonic oscillators or the non-trivial Toda lattice[1]). Thus, one has to use a probabilistic approach as the one based on the methods of classical statistical mechanics.

Let $\rho(q,p,t)$ stand for the density of states in phase space. Liouville's theorem says that

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^{3N} \left(\frac{\partial \rho}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} \right) = 0 \tag{2.2}$$

[1] A thorough discussion about the Toda lattice and its integrable structure can be found in [6].

Density of states of the form $\rho(q,p,t) = \rho(H(q,p),t)$ imply $\partial\rho/\partial t = 0$, which means that they are stationary. Examples are the canonical distribution $\rho \sim e^{-\beta H}$ and the microcanonical distribution $\rho \sim \delta(H-E)$.

2.1 Microcanonical Ensemble

A point in phase space is called an accessible state if the system can evolve to this configuration. The main assumption of statistical mechanics is the **equal a priori probability postulate**, which states that all accessible states are equally likely to occur, *i.e* ρ is constant on the hypersurface of constant energy in phase space. This is called the microcanonical ensemble.

First, consider the volume $\Sigma(E)$ of the region $H < E$, given by the formula

$$\Sigma(E) = \frac{1}{h^{3N}} \int\limits_{H<E} d^{3N}q \; d^{3N}p \tag{2.3}$$

where the $1/h^{3N}$ factor is added in order to make $\Sigma(E)$ dimensionless. The area $\omega(E)$ of the $H = E$ hypersurface is then given by

$$\omega(E) = \frac{\partial\Sigma(E)}{\partial E} \tag{2.4}$$

Now, we can define the microcanonical entropy as

$$S(E,V) := \ln \omega(E) \tag{2.5}$$

where we have set the Boltzmann constant $k_B = 1$. From (2.5), we can obtain thermodynamic parameters like temperature T and pressure P by means of the definitions

$$\frac{1}{T} := \frac{\partial S}{\partial E} \qquad P := T\frac{\partial S}{\partial V} \tag{2.6}$$

2.1.1 Classical Ideal Gas

Let us consider some examples. The classical ideal gas is defined by the Hamiltonian

$$H = \frac{1}{2m} \sum_{i=1}^{3N} p_i^2 \tag{2.7}$$

Then, we get from (2.3)

$$\Sigma(E) = \frac{1}{h^{3N}} \int\limits_{\sum_i p_i^2 < 2mE} d^{3N}q \; d^{3N}p = \left(\frac{V}{h^3}\right)^N \Omega_{3N}(\sqrt{2mE}) \tag{2.8}$$

where $\Omega_n(R)$ denotes the volume of the $n-$sphere of radius R. In order to perform this calculation, we use the trick described in [7]. By means of dimensional arguments, we have

$$\Omega_n(R) = C_n R^n \rightarrow d\Omega_n = nC_n R^{n-1} dR \tag{2.9}$$

where C_n is a constant number. Our aim is to determine C_n.

Now, the trick is to consider the integral

$$\int_{\mathbb{R}^n} e^{-|\vec{x}|^2} d\vec{x} = \left(\int_{\mathbb{R}} e^{-x^2} dx \right)^n = \pi^{n/2} \tag{2.10}$$

In spherical coordinates, this integral becomes

$$\int_0^\infty e^{-R^2} n C_n R^{n-1} dR = \frac{n}{2} C_n \Gamma(n/2) = C_n \Gamma(n/2+1) \tag{2.11}$$

Thus, we have

$$C_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)} \Rightarrow \ln C_n \sim \frac{n}{2} \ln \pi - \frac{n}{2} \ln \left(\frac{n}{2} \right) + \frac{n}{2} \quad \text{as } n \to \infty \tag{2.12}$$

Substituting these results in (2.9) and (2.8) and taking into account formula (2.4) and the definition (2.5) gives us the final result at the leading order in N

$$S(E,V) = N \ln \left[V \left(\frac{4\pi m}{3h^2} \frac{E}{N} \right)^{3/2} \right] + \frac{3}{2} N \tag{2.13}$$

2.2 Canonical Ensemble

Besides microcanonical ensemble, there exists an another formulation of statistical mechanics, which is the canonical ensemble. To define canonical ensemble, consider a system S_1 lying inside a heat bath, which we call system S_2. We will assume system S_2 to be very large when compared to the system S_1, i.e.

$$N_2 \gg N_1 \qquad \overline{E_2} \gg \overline{E_1}$$

where the overbar stands for average. Let us also denote the total energy by E, i.e. $E \approx \overline{E_1} + \overline{E_2}$ where we have neglected the interaction energy between the two systems.

Finding the system S_1 in a microstate (q_1, p_1) within $dq_1 dp_1$ is proportional to $\Gamma_2(E - E_1) dq_1 dp_1$, where E_1 is now a fluctuating quantity. Thus,

$$\rho(q_1, p_1) \sim \Gamma_2(E - E_1) \tag{2.14}$$

Now, by definition of entropy we have (recall $k_B = 1$)

$$\ln \Gamma_2(E - E_1) = S_2(E - E_1) \approx S_2(E) - \frac{E_1}{T} \tag{2.15}$$

Hence, as a final result we get

$$\rho(q,p) \sim e^{-H(p,q)/T} \tag{2.16}$$

Adding a dimensional factor and the combinatorial factor $1/N!$ in order to remove the Gibbs paradox, we call the normalization factor of this probability density as the **canonical partition function**[2]:

$$Z_N(V,T) = \int \frac{d^{3N}q\, d^{3N}p}{N!\, h^{3N}} e^{-\beta H(q,p)} := e^{-\beta A(V,T)} \tag{2.17}$$

where $\beta = 1/T$. To understand the physical meaning of $A(V,T)$, differentiate both sides of (2.17) with respect to β to obtain

$$\overline{E}(V,T) = A(V,T) - T\frac{\partial A}{\partial T}\bigg|_V \tag{2.18}$$

From there we can see that $A(V,T)$ corresponds to the Helmholtz free energy in thermodynamics.

2.2.1 Classical Ideal Gas

Let us repeat the calculation of the ideal gas in the canonical ensemble. The Hamiltonian of the ideal gas is

$$H = \frac{1}{2m}\sum_{i=1}^{3N} p_i^2 \tag{2.19}$$

Then, we have the canonical partition function

$$\begin{aligned} Z_N &= \frac{1}{N!\, h^{3N}} \int d^{3N}p\, d^{3N}q\; e^{-\frac{\beta}{2m}\sum_i p_i^2} \\ &= \frac{1}{N!}\left(\frac{V}{h^3}\right)^N \left(\int_{-\infty}^{\infty} dp\, e^{-\beta p^2/2m}\right)^{3N} \\ &= \frac{1}{N!}\left[\frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2}\right]^N \end{aligned} \tag{2.20}$$

Then, we can find the Helmholtz free energy as (below, again $k_B = 1$)

$$A(V,T) = -T\ln Z_N \approx -NT\left[\ln\left\{\frac{V}{N}\left(\frac{2\pi m T}{h^2}\right)^{3/2}\right\} + 1\right] \tag{2.21}$$

From this, we can find various properties of ideal gas. For instance

$$P = -\frac{\partial A}{\partial V} = \frac{NT}{V} \tag{2.22}$$

[2] As can be seen, for the convergence of the integral in the (2.17), in most cases one should assume $\beta > 0$, *i.e.* $T > 0$. If this is not the case, then the canonical ensemble may be undefined.

the well-known ideal gas law, and the expression of the entropy for the ideal gas in the canonical ensemble

$$S = -\frac{\partial A}{\partial T} = N \ln \left[\frac{V}{N} \left(\frac{4\pi m}{3h^2} \frac{E}{N} \right)^{3/2} \right] + \frac{5}{2} N \tag{2.23}$$

From (2.18), we can also find the average energy $\overline{E}$ as

$$\overline{E} = \frac{3}{2} N T \tag{2.24}$$

the well known equipartition law.

2.3 Equivalence of Microcanonical-Canonical Ensembles

In the microcanonical ensemble, the system is isolated and has a constant energy. In contrast, in the canonical ensemble the system interacts with the environment and hence exhibits energy fluctuations. Nevertheless, we can show that these fluctuations are negligible, and hence the two ensembles are equivalent.

To show this, consider the expression

$$\overline{E} = \frac{\int dpdq \; H e^{-\beta H}}{Z} \tag{2.25}$$

implying

$$\int dpdq \left(\overline{E} - H \right) e^{\beta(A-H)} = 0 \tag{2.26}$$

Differentiating both sides with respect to β in (2.26) gives

$$\Delta H = \sqrt{T^2 C_V} \Rightarrow \frac{\Delta H}{H} \sim \frac{1}{\sqrt{N}} \ll 1 \tag{2.27}$$

We can also get the same result as follows: By definition of $\omega(E)$ we have

$$Z_N = \frac{1}{N! \, h^{3N}} \int dpdq \, e^{-\beta H} = \int_0^\infty dE \; \omega(E) e^{-\beta E} = \int_0^\infty dE \; e^{\beta(TS(E)-E)} \tag{2.28}$$

The maximum of the integrand of (2.28) occurs at $E = E^*$ where

$$T \left(\frac{\partial S}{\partial E} \right)_{E=E^*} = 1 \qquad \left(\frac{\partial^2 S}{\partial E^2} \right)_{E=E^*} < 0 \tag{2.29}$$

The first equality in (2.29) says that $E^* = \overline{E}$ the average energy while the second says that the heat capacity C_V satisfies $C_V > 0$, an expected thermodynamic inequality[3].

[3]If $C_V < 0$, then the two ensembles may be non-equivalent. This phenomenon is called the **ensemble inequivalence**.

Now, consider the Taylor expansion

$$TS(E) - E = TS(\overline{E}) - \overline{E} - \frac{1}{2TC_V}(E - \overline{E})^2 + \dots \tag{2.30}$$

Substituting this expansion to (2.28) gives

$$Z_N \approx e^{\beta(TS(\overline{E}) - \overline{E})} \int_0^\infty dE\, e^{-(E-\overline{E})^2/2T^2C_V} \tag{2.31}$$

Notice that the integrand in (2.31) is a Gaussian distribution with $\Delta E = \sqrt{T^2 C_V}$.

Performing the Gaussian integral in (2.31) gives

$$A \approx \overline{E} - TS - \frac{1}{2}T \ln C_V \tag{2.32}$$

Thus, the error from the usual thermodynamic relation $A = \overline{E} - TS$ is $T \ln C_V/2 \sim O(\ln N)$, which is negligible. Hence, we showed that the two ensembles are equivalent.

Chapter 3

Dynamical Approach to Thermodynamics

As we discussed in the previous chapter, the aim of statistical mechanics is to understand the mechanics of large systems by means of statistical methods. Thus, it may be expected that the thermodynamic quantities like entropy and temperature should have some dynamical interpretation. This is indeed true, at least under the assumption of **ergodicity**. If a system is ergodic, then the dynamical average taken over time is equal to the phase space average, *i.e.* we have

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T f(q(t),p(t))\,dt = \int_{\mathcal{M}} dq\,dp\,f(q,p) \tag{3.1}$$

where the time average on the l.h.s. is known to exist thanks to Birkhoff theorem[1]. In there, $(q(t),p(t))$ is the dynamical evolution of a generic initial condition $(q(0),p(0))$, which is determined by the Hamilton equations, and $\mathcal{M}$ stands for the phase space of the system. Intuitively, ergodicity can be understood to mean that an ergodic system spends equal amounts of time in each accessible state in the phase space, hence the assumption of ergodicity is closely related to equal a priori probability postulate.

Although truly ergodic systems might theoretically appear rare, the empirical success of statistical mechanics implies that many physical systems effectively exhibit ergodic behavior.

For more information about relation between ergodicity and foundations of statistical mechanics, one may consult to the book [9], which also contains a discussion of formula (3.1). See also Appendix K.

[1] There are some recent claims that this theorem was first discovered by Von Neumann; see [8] for a discussion.

3.1 The Mechanical Interpretation of Entropy

Entropy is known to be unchanged under reversible adiabatic processes. A similar quantity in classical mechanics is the action integral or the so called adiabatic invariant, which is defined as the symplectic volume

$$I = \int p\, dq \tag{3.2}$$

See [10] for a discussion of adiabatic invariance. Hence, we can expect that there is a relation between the entropy and the adiabatic invariant, and this is indeed true as we will show in this section.

Let us start with an example, consider a 1-D mass m, restricted to the interval $(0, L)$. Its dynamics can be described by the Hamiltonian

$$H(q, p; L) = p^2/2m + U_{box}(q; L) \tag{3.3}$$

where we have the constraining potential of the box

$$U_{box}(q; L) = \begin{cases} 0 & x \in (0, L) \\ \infty & \text{Otherwise} \end{cases} \tag{3.4}$$

The dynamics of this particle is trivial; the particle moves back and forth inside the interval with constant speed v_0.

Let us consider the force on the, say, right wall. Per collision, the momentum change of the particle is given by $\Delta p = 2mv_0$ while the time between each collision is equal to $\Delta t = 2L/v_0$. Thus, the second law of Newton says

$$F = \frac{\Delta p}{\Delta t} = \frac{mv_0^2}{L} = \frac{2E}{L} \tag{3.5}$$

Now, consider the first law of thermodynamics

$$\delta Q = dE - \delta W = dE + F dL = dE + \frac{2E}{L} dL \tag{3.6}$$

In (3.6), δQ can be made exact by multiplying with an integrating factor $k_B/2E$, which is traditionally defined as the inverse temperature $1/T$:

$$k_B T = 2E \Rightarrow \frac{\delta Q}{T} = dS = d\left(k_B \ln\left(\sqrt{E}L\right)\right) \Rightarrow S = k_B \ln\left(\sqrt{E}L\right) + C \tag{3.7}$$

Also, notice that we have $FL = k_B T$, a one dimensional analogue of the ideal gas law.

Notice that $S(E, L)$ is the adiabatic invariant of the system. This can be demonstrated as follows:

$$I = \oint pdq = \iint dp\, dq = mv_0 \times 2L \sim \sqrt{E}L \Rightarrow S \sim k_B \ln I \tag{3.8}$$

This analogy holds more generally, which is the conclusion of the following theorem[2]:

Theorem 3.1.1. *(Helmholtz Theorem) Consider a 1-D system with U-shaped potential $U(q;\lambda)$ with an external parameter λ. Define the quantities*

$$k_B T = \langle p^2/m \rangle_t \qquad \Lambda = -\langle \partial H/\partial\lambda \rangle_t \tag{3.9}$$

where the time averages $\langle f \rangle_t$ are taken over one period.

If we also define the entropy as the logarithm of the phase space area, i.e. $S = k_B \ln I$, we get

$$dS = \frac{dE + \Lambda d\lambda}{T} \tag{3.10}$$

As an example, consider the harmonic oscillator

$$H(x, p; \lambda) = p^2/2m + m\lambda^2 x^2/2 \tag{3.11}$$

The entropy calculation gives

$$S = k_B \ln I \sim k_B \ln{(E/\lambda)} \Rightarrow k_B T = E \quad \Lambda = -E/\lambda \tag{3.12}$$

Now, let us look at the time average more closely. If τ is the period of the motion, we have

$$\langle f \rangle_t = \frac{1}{\tau} \int_0^\tau f(q(t), p(t))\, dt \tag{3.13}$$

Now, using the expression $dt = mdq/p(q)$ we get

$$\langle f \rangle_t = \frac{2m}{\tau} \int \frac{dq}{p(q)} f(q, p(q)) \tag{3.14}$$

Moreover, using the identity

$$\int dp\ \delta(p^2/2m + U(q) - E) = \frac{2m}{p(q)} \tag{3.15}$$

and the fact that $\tau = \int dt$ leads us to the expression

$$\langle f \rangle_t = \frac{\iint dpdq\ \delta(p^2/2m + U(q) - E)\ f(q, p)}{\iint dpdq\ \delta(p^2/2m + U(q) - E)} \tag{3.16}$$

the usual formula for the microcanonical ensemble. In particular, we showed that the systems satisfying the hypothesis of the Helmholtz theorem are ergodic.

[2]For the proof consult the book [11], which is also the main source of this section.

3.2 The Rugh Temperature

In this section, we will cover the differential geometric aspects of the derivation of thermodynamics from classical mechanics.

A classical mechanical system can be defined in terms of a symplectic manifold $(\mathcal{M}, \omega)$ where $\mathcal{M}$ is a smooth manifold endowed with a symplectic form ω. From there, one can define the Liouville volume form $m = \omega^{\wedge n}$. See also [12].

We will focus on the symplectic manifold $(\mathbb{R}^{2n}, \sum_i dq_i \wedge dp_i)$, where the Liouville volume form m becomes the Lebesgue measure.

To define a dynamics, we consider a function $H \in C^\infty(\mathcal{M})$ which we can use to define a vector field X_H, called the Hamiltonian vector field, by means of the Hamilton's equations.

A well-known theorem states that the flow of X_H, which we will denote as g^t, preserves m and the level set $\Sigma(E) = H^{-1}(E)$. As a result, the area form of the $\Sigma(E)$ will also be preserved.

$$\mu_E = \text{Restriction of } m \text{ to } \Sigma(E) = m\delta(H - E) \tag{3.17}$$

Then, we can define the phase space averages on $\Sigma(E)$ as

$$\langle \phi \rangle_E = \frac{\int_{\mathcal{M}} m\, \delta(H - E)\phi}{\int_{\mathcal{M}} m\, \delta(H - E)} \tag{3.18}$$

Now, we can define the Boltzmann entropy and microcanonical temperature as

$$S(E) = \ln \left(\int_{\mathcal{M}} m\, \delta(H - E) \right) \qquad T(E) = \frac{\partial S(E)}{\partial E} \tag{3.19}$$

Theorem 3.2.1. *Assume that $\Sigma(E)$ is a smooth surface and X is a vector field defined in a neighborhood of $\Sigma(E)$, satisfying $\nabla H \cdot X = 1$. Then, we have*

$$\frac{\partial}{\partial E} \int_{\mathcal{M}} m\, \delta(H - E)\, \phi = \int_{\mathcal{M}} m\, \delta(H - E)\, div(\phi X) \tag{3.20}$$

Proof. First, notice that we have the relations

$$\int_{\mathcal{M}} m\delta(H - E)\phi = \int_{H=E} \frac{d\Sigma}{||\nabla H||}\phi \Rightarrow \int_{H \leq E} m\phi = \int_{-\infty}^{E} du \int_{H=u} \frac{d\Sigma}{||\nabla H||}\phi$$

Now, using contractions and differential forms, we obtain

$$m\delta(H - E) = \mu_E = i_X m \rightarrow d(\mu_E \phi) = d(i_{\phi X} m) = div(\phi X)m$$

Hence, by Stokes theorem

$$\int_{H=E} \mu_E \phi = \int_{H \leq E} d(\mu_E \phi) = \int_{H \leq E} m\, div(\phi X) = \int_{-\infty}^{E} du \int_{H=u} \mu_E\, div(\phi X)$$

Then, we get the desired result (3.20) by differentiating the first and the final formulas above with respect to E.

$\square$

In particular, let $\phi = 1$ and $X = \nabla H/||\nabla H||^2$. Then, the formula (3.20) combined with ergodicity implies

$$\frac{1}{T(E)} = \lim_{t\to\infty} \frac{1}{t} \int_0^t d\tau\ \Phi(x(\tau)) \quad \text{where} \quad \Phi = \nabla \cdot \left(\nabla H/||\nabla H||^2\right) \tag{3.21}$$

for almost all $x(0) \in \Sigma(E)$. This definition of temperature is denoted in the literature ([13],[14]) as the *Rugh temperature*.

For instance, let us consider the classical ideal gas $H = \sum p_i^2/2m$. A direct computation gives

$$\Phi = \frac{3N-2}{2H} \Rightarrow \frac{1}{t}\int_0^t d\tau\ \Phi(x(\tau)) = \frac{3N-2}{2H} \tag{3.22}$$

Equating the last formula above to $1/T$ gives the usual energy expression for the ideal gas in the large N limit

$$H \approx \frac{3N}{2}T. \tag{3.23}$$

More generally, consider the system

$$H = \sum_i p_i^2/2m + U(q) \tag{3.24}$$

Here, we can choose X as

$$X = \left(\vec{0}, \frac{\vec{p}}{2K(p)}\right) \tag{3.25}$$

Then, (3.20) gives us the result

$$\frac{1}{T} = \frac{3N-2}{2}\left\langle \frac{1}{K(p)} \right\rangle \tag{3.26}$$

Notice that the last expression differs from the usual definition of the temperature, which is given as

$$T^* = \frac{2}{3N}\left\langle K(p) \right\rangle \tag{3.27}$$

Indeed, we have

$$\frac{T^*}{T} = \frac{3N-2}{3N}\left\langle K(p) \right\rangle \left\langle \frac{1}{K(p)} \right\rangle \tag{3.28}$$

Hence, in the $N \to \infty$ limit where the fluctuations in the kinetic energy $K(p)$ become negligible, we get $T = T^*$.

3.3 Mean Field ϕ^4 Theory

As an example to the Rugh temperature considered in the previous section, consider the system defined by the mean field ϕ^4 Hamiltonian

$$H = \sum_{i=1}^{N} \left(\frac{p_i^2}{2} - \frac{1}{4} q_i^2 + \frac{1}{4} q_i^4 \right) - \frac{\alpha}{4N} \sum_{i,j}^{N} q_i q_j \tag{3.29}$$

This model has been introduced in [15]. It represents particles sitting in a double well potential $V(x) \sim x^4 - x^2$ which are interacting all-to-all. A global minimum of $V(x)$, i.e. a vacuum state of the system is $q_i = 1/\sqrt{2}$, and in this section we will assume $\alpha \ll 1$ which allows us to consider small oscillations around this vacuum:

$$q_i = \frac{1}{\sqrt{2}} + \epsilon_i \tag{3.30}$$

Also, we will assume that the particles start from the vacuum with zero momenta, i.e. $\epsilon_i(0) = 0 = p_i(0)$.

Substituting this into the Hamiltonian and ignoring the terms $O(\epsilon_i^3)$ gives

$$H \simeq \sum_{i=1}^{N} \left(\frac{p_i^2}{2} + \frac{1}{2} \epsilon_i^2 - \frac{\alpha}{2\sqrt{2}} \epsilon_i \right) - \frac{\alpha}{4N} \left(\sum_{i=1}^{N} \epsilon_i \right)^2 \tag{3.31}$$

As usual in classical mechanics, we can diagonalize this Hamiltonian by considering an orthogonal coordinate transformation $(\epsilon_i, p_i) \mapsto (Q_i, P_i)$:

$$\widetilde{Q}_k = \sum_{i=1}^{N} R_{ki} \epsilon_i \tag{3.32}$$

$$\widetilde{P}_k = \sum_{i=1}^{N} R_{ki} p_i \tag{3.33}$$

where R is an orthogonal matrix, $R^T R = I$, and we choose $R_{1j} = 1/\sqrt{N}$ for $j = 1, 2, \ldots, N$, implying

$$\sum_{i=1}^{N} \epsilon_i = \sqrt{N} \widetilde{Q}_1 \tag{3.34}$$

Also, since R is an orthogonal matrix we have

$$\sum_{i=1}^{N} \epsilon_i^2 = \sum_{k=1}^{N} \widetilde{Q}_k^2 \quad \text{and} \quad \sum_{i=1}^{N} p_i^2 = \sum_{k=1}^{N} \widetilde{P}_k^2 \tag{3.35}$$

Now, substituting all of these into the Hamiltonian H gives

$$H = \frac{1}{2}\sum_{i=1}^{N}\widetilde{P}_i^2 + \frac{1}{2}\sum_{i=1}^{N}\widetilde{Q}_i^2 - \frac{\alpha\sqrt{N}}{2\sqrt{2}}\widetilde{Q}_1 - \frac{\alpha}{4}\widetilde{Q}_1^2 \tag{3.36}$$

Now, all modes except the $k = 1$ mode are in the harmonic oscillator form, while the $k = 1$ mode can be turned to the harmonic oscillator form by completing the square. Hence, we define

$$Q_1 = \widetilde{Q}_1 - \frac{\alpha\sqrt{2N}}{2(2-\alpha)} \quad \text{and} \quad Q_k = \widetilde{Q}_k \text{ for } k > 1 \tag{3.37}$$

The result is

$$H = \frac{P_1^2}{2} + \frac{1}{2}\omega_1^2 Q_1^2 + \sum_{i>1}^{N}\left(\frac{P_i^2}{2} + \frac{1}{2}Q_i^2\right) \tag{3.38}$$

where $\omega_1^2 = 1 - \alpha/2$.

Also, in these new coordinates our initial conditions become $Q_i(0) = 0 = P_i(0)$ for $i > 1$ and

$$Q_1(0) = -\frac{\alpha\sqrt{2N}}{2(2-\alpha)} \qquad P_1(0) = 0 \tag{3.39}$$

Now, we can easily integrate the equations of motion for this system. We get $Q_i(t) = 0$ for $i > 1$ and

$$Q_1(t) = -\frac{\alpha\sqrt{2N}}{2(2-\alpha)}\cos(\omega_1 t) \tag{3.40}$$

Now, consider $\Phi = \nabla \cdot \left(\nabla H/||\nabla H||^2\right)$. Explicitly calculating this quantity and averaging it over one period gives

$$\langle\Phi\rangle = \frac{8(N-1)}{N\alpha^2}\sqrt{4-2\alpha} \Rightarrow T = \frac{1}{8}\frac{N}{N-1}\frac{\alpha^2}{\sqrt{4-2\alpha}} \tag{3.41}$$

We can also show this numerically, by running a simulation that is given in Appendix A. The result is as follows:

α	Simulated T (Exact Dynamics)	T (Rugh Temperature)
0.1	0.000629739	0.000663348
0.09	0.000511569	0.000535904
0.08	0.000405414	0.000422326
0.07	0.000311308	0.000322504
0.06	0.000229346	0.000236331
0.05	0.000159679	0.000163697
0.04	0.000102441	0.000104499
0.03	0.0000577625	0.000058631
0.02	0.0000257335	0.0000259924
0.01	6.45002×10^{-6}	6.48174×10^{-6}

Figure 3.1: Comparison of numerical calculations with (3.41)

The numerical results generated by the code in Appendix A confirm that the analytical calculation is highly accurate for small α, as expected.

We can also look for the dependence of magnetization $m = \langle | \sum_i q_i | \rangle$ and temperature T to the energy via the code in Appendix B.

It gives the following result:

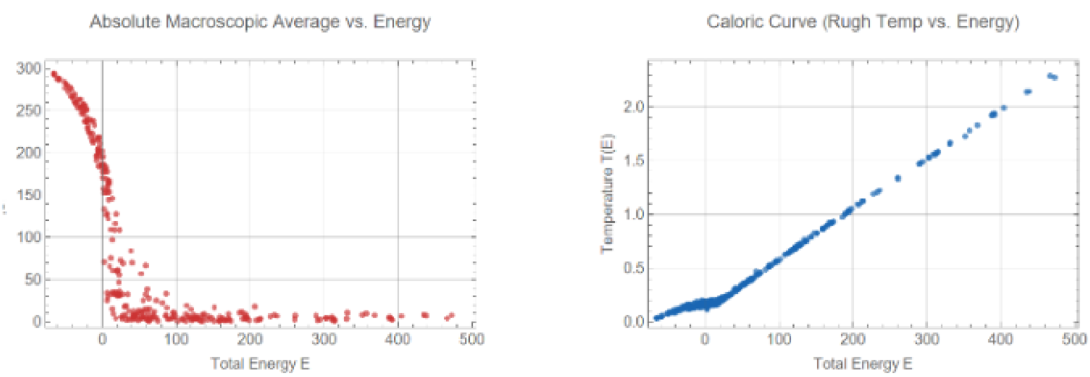

Figure 3.2: Magnetization and Temperature vs. Energy

There are two interesting results in Figure 3.2:

- For $E > 0$, the magnetization drops to zero, which clearly indicates a phase transition.
- For $E > 0$, the temperature depends linearly on energy, whereas for $E < 0$, this dependence becomes non-linear.

These two observations can be naturally explained through physical arguments. When $E < 0$, each particle is energetically trapped within one of the two local minima of the double-well potential $V(x) \sim x^4 - x^2$. Conversely, for $E > 0$, the particles possess sufficient energy to escape these wells and explore the full configuration space. This confinement mechanism explains why the magnetization is non-zero when $E < 0$ and vanishes for $E > 0$.

Furthermore, when the net magnetization is zero, the collective mean-field interaction effectively decouples. Consequently, the total Hamiltonian simplifies to a sum of independent, single-particle Hamiltonians. In this non-interacting regime ($E > 0$), the generalized equipartition theorem applies, directly giving the linear proportionality $T \sim E$.

3.4 Revisiting Mean Field ϕ^4 Theory

Recall that the mean field ϕ^4 theory is defined by means of the Hamiltonian (3.29) (now $\alpha = 1$)

$$H = \sum_{i=1}^{N} \left(\frac{p_i^2}{2} - \frac{1}{4} q_i^2 + \frac{1}{4} q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^{N} q_i q_j \tag{3.42}$$

where the only conserved quantity is energy. In the previous chapter we have studied the dynamics where only the energy was constrained and the magnetization was assuming the most probable value. Let us now also constrain the dynamics in such a way that the magnetization

$$m = \frac{1}{N} \sum_{i=1}^{N} q_i \equiv \frac{M}{N} \tag{3.43}$$

is now exactly conserved. There are several ways one can realize this constraint: *i)* Removal of one dynamical variable; *ii)* Use of Dirac brackets; *iii)* Addition of Lagrange multiplier. The last method will be described in Sec. 3.5.3.

3.4.1 Removal of One Dynamical Variable

One way to remove one variable consists the formula $\sum_i q_i = M$ and its derivative $\sum_i p_i = \sum_i \dot{q}_i = 0$ in order to eliminate q_N and p_N. The resulting Hamiltonian containing only the remaining variables is

$$\begin{aligned} H = {} & \frac{1}{2} \sum_{i=1}^{N-1} p_i^2 - \frac{1}{2N} \sum_{i,j}^{1,N-1} p_i p_j + \sum_{i=1}^{N-1} \left(-\frac{1}{4} q_i^2 + \frac{1}{4} q_i^4 \right) \\ & - \frac{1}{4} \left(M - \sum_{i=1}^{N-1} q_i \right)^2 + \frac{1}{4} \left(M - \sum_{i=1}^{N-1} q_i \right)^4 - \frac{M^2}{4N} \end{aligned} \tag{3.44}$$

The corresponding equations of motion are

$$\dot{q}_k = p_k - \frac{1}{N} \sum_{j=1}^{N-1} p_j \tag{3.45}$$

$$\dot{p}_k = \frac{1}{2} q_k - q_k^3 - \frac{1}{2} \left(M - \sum_{j=1}^{N-1} q_j \right) + \left(M - \sum_{j=1}^{N-1} q_j \right)^3 \tag{3.46}$$

The code which integrates the equations of motions is reported in Appendix C. In fig. 3.1 we show the time evolution of both the energy and magnetization, proving that the initial value is well-conserved[3].

[3] It has to be noted that while energy fluctuates around a mean value, magnetization always increases in time although in a limited numerical range.

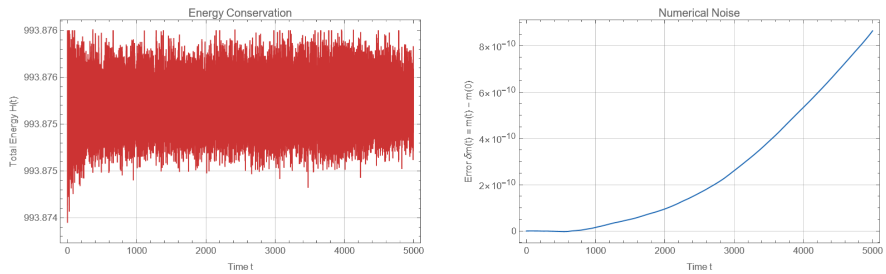

Figure 3.3: Left: Time Evolution of Energy Right: Time Evolution of Magnetization

The reason why energy fluctuates without showing a drift is due to the use of a symplectic integrator, a numerical algorithm designed to solve the equations of motion for Hamiltonian systems while strictly preserving their underlying geometric structure [16]. Due to this, there is no artificial energy drift over time in symplectic integrators, other numerical algorithms like Euler or Runge-Kutta will generate drift in the energy. Higher-order symplectic methods are also available, like the Yoshida algorithm [17]. However, for the purpose of the simulations done in this project, it was enough to use the Verlet algorithm, which gives energy fluctuations of the order $(dt)^2$ where dt is the time step.

3.4.2 Use of Dirac Brackets

We can also get the equations of motions (3.45) by using the Dirac-Bergmann formalism[4]. To use this formalism, first consider our starting Hamiltonian

$$H_0 = \sum_{i=1}^{N} \left(\frac{p_i^2}{2} - \frac{1}{4} q_i^2 + \frac{1}{4} q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^{N} q_i q_j \tag{3.47}$$

and our constraint

$$\phi_1 \approx \frac{1}{N} \sum_{i=1}^{N} q_i - m$$

and construct the Hamiltonian $H_T = H_0 + \lambda \phi_1$. The consistency condition $\dot{\phi}_1 = \{\phi_1, H_T\} \approx 0$ gives the secondary constraint

$$\phi_2 = \frac{1}{N} \sum_{i=1}^{N} p_i \approx 0$$

Now, the new consistency condition $\dot{\phi}_2 = \{\phi_2, H_T\} \approx 0$ gives

$$\lambda = \sum_{i=1}^{N} (q_i - q_i^3)$$

[4]Dirac himself has a great lecture course [18] on this topic.

Hence, the Dirac-Bergmann algorithm terminates here. Our resulting Hamiltonian is

$$H = \sum_{i=1}^{N} \left(\frac{p_i^2}{2} - \frac{1}{4}q_i^2 + \frac{1}{4}q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^{N} q_i q_j + \sum_{i=1}^{N} (q_i - q_i^3) \left(\frac{1}{N} \sum_{i=1}^{N} q_i - m \right) \tag{3.48}$$

We can also deduce the Dirac bracket, given as

$$\{q_i, q_j\}_D = \{p_i, p_j\}_D = 0 \qquad \{q_i, p_j\}_D = \delta_{ij} - \frac{1}{N} \tag{3.49}$$

By using this Dirac bracket, we can obtain equations of motion equivalent under a canonical transformation to (3.45).

In Appendix D, we present a code which integrates the constrained equations of motion. Running the code, one can compute the dependence of temperature to energy and check the conservation of magnetization. See fig. 3.4.

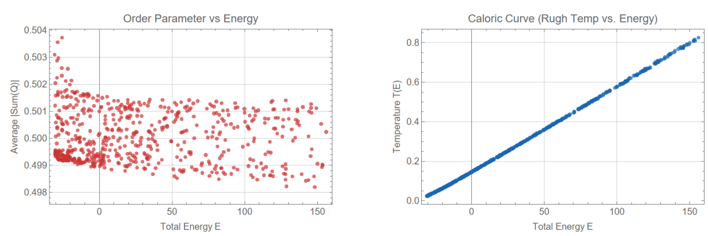

Figure 3.4: Left: We show the small fluctuations of magnetization, proving it is well-conserved for the initial value $m = 0.5$. Right: We report the dependence of temperature to energy, which is almost a line.

We can also compare the different values of m via the second program in Appendix D. As we can see, the heat capacity is proportional to the magnetization m.

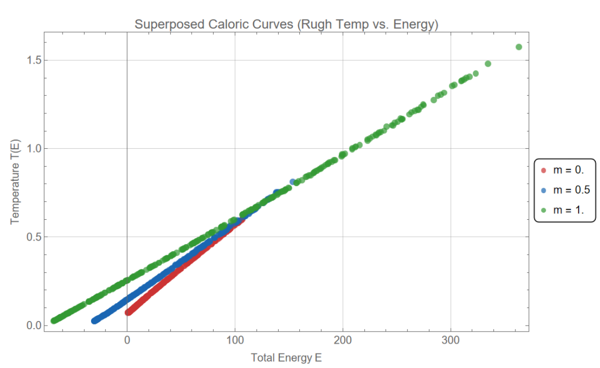

Figure 3.5: Caloric curves for different values of magnetization

3.5 Systems with Two Conserved Quantities

Let us consider a system with $N+2$ phase space variables $(x^1, \dots, x^N, x^{N+1}, x^{N+2}) := (\vec{x}, x^{N+1}, x^{N+2})$, such that there exist two conserved quantities, namely the Hamiltonian H and another conserved quantity V.

The system lies in the set

$$\mathcal{M} = \{(\vec{x}, x^{N+1}, x^{N+2}) \in \mathbb{R}^{N+2} \,|\, H = E,\ V = u\} \tag{3.50}$$

We can define the surface entropy as

$$S(E) = \ln \int d^N\vec{x}\, dx^{N+1} dx^{N+2}\ \delta(H - E)\delta(V - u) \tag{3.51}$$

Generically, $\mathcal{M}$ is a smooth submanifold and for $x_0 \in \mathcal{M}$, $\nabla H(x_0) \neq 0 \neq \nabla V(x_0)$. Since H and V are functionally independent, up to a reordering of indices we can assume $\frac{\partial H}{\partial x^{N+1}} \frac{\partial V}{\partial x^{N+2}} - \frac{\partial V}{\partial x^{N+1}} \frac{\partial H}{\partial x^{N+2}} \neq 0$ in a neighborhood of $x_0 \in \mathcal{M}$. Now, by a partition of unity argument, we can assume this condition holds globally.

Let us parametrize $\mathcal{M}$ as $(\vec{x}, g(\vec{x}), h(\vec{x}))$. Then, the induced metric and its determinant are given as

$$\eta_{ij} = \delta_{ij} + \partial_i g\, \partial_j g + \partial_i h\, \partial_j h \Rightarrow \eta = 1 + \sum_i^N \left[(\partial_i g)^2 + (\partial_i h)^2\right] + \sum_{i<j}^N (\partial_i g\, \partial_j h - \partial_j g\, \partial_i h)^2 \tag{3.52}$$

Also, we can express the derivatives $\partial_\alpha g, \partial_\alpha h$ as

$$\partial_\alpha g = \left[\partial_{N+2} V\, \partial_\alpha H - \partial_{N+2} H\, \partial_\alpha V\right]/D \qquad \partial_\alpha h = \left[\partial_{N+1} H\, \partial_\alpha V - \partial_{N+1} V\, \partial_\alpha H\right]/D \tag{3.53}$$

where $D = \partial_{N+1} H\, \partial_{N+2} V - \partial_{N+1} V\, \partial_{N+2} H \neq 0$. Now, we can obtain the volume form

$$d\tau = \sqrt{\eta}\, d^N\vec{x} = \frac{W}{D} d^N\vec{x} \tag{3.54}$$

where

$$W = \sqrt{\sum_{i<j}^{N+2} (\partial_i H\, \partial_j V - \partial_i V\, \partial_j H)^2} \tag{3.55}$$

Now, defining $y^1 = H(\vec{x}, x^{N+1}, x^{N+2}) - E$ and $y^2 = V(\vec{x}, x^{N+1}, x^{N+2}) - u$ gives

$$S(E) = \ln \int d^N\vec{x}\ dy^1 dy^2\ \delta(y^1)\delta(y^2)\ \det\left(\frac{\partial(x^{N+1}, x^{N+2})}{\partial(y^1, y^2)}\right) = \ln \int_{\mathcal{M}} \frac{d\tau}{W} \tag{3.56}$$

3.5.1 Federer-Laurance Derivation Formula

To simplify the calculations for systems with 2 conserved quantities, we can use the following generalization of the Federer-Laurance derivation formula: First, define the normal vectors

$$n^H = \frac{\nabla H}{||\nabla H||} \quad n^V = \frac{\nabla V}{||\nabla V||} \tag{3.57}$$

Then, define the projection of n^H to n^V and the corresponding unit vector as

$$\xi = n^H - (n^H \cdot n^V)n^V \qquad n^\xi = \frac{\xi}{||\xi||} \tag{3.58}$$

Now, define the operator $A(\#)$ as

$$A(\#) = \frac{1}{\nabla H \cdot \#} \left[\nabla(\# n^\xi) - \# n^V \cdot (n^V \cdot \nabla)(n^\xi)\right] \tag{3.59}$$

Finally, we have the Federer-Laurance derivation formula:

$$\frac{\partial^k}{\partial E^k} \int_{\mathcal{M}} f \, d\tau = \int_{\mathcal{M}} A^k(f) \, d\tau \tag{3.60}$$

This formula can be used to calculate various physical quantities like temperature, heat capacity etc.

As an example[5], consider an N particle system governed by the Hamiltonian

$$H = \frac{1}{2} \sum_m (p_m^2 + q_m^2)^2 - \sum_m (p_m p_{m+1} + q_m q_{m+1}) \tag{3.61}$$

with periodic boundary conditions. This system, besides Hamiltonian H, has the conserved quantity

$$V = \sum_m (p_m^2 + q_m^2)/2 \tag{3.62}$$

To find the ground state of this system, we need to use the Lagrange multiplier method, $\delta(H - \lambda V) = 0$. This gives the stable ground state

$$q_i = \sqrt{2V/N} := q_0 \qquad p_i = 0 \tag{3.63}$$

For low energies around the ground state, our formula gives $T = 0$. If we apply the formula for system with only conserved quantity as a Hamiltonian, we get an inaccurate result

$$T = \frac{1}{2}(q_0^2 - 1)^2 \tag{3.64}$$

[5] This example is taken from [19].

3.5.2 Linear Constraints

As an example, let us assume that our second conserved quantity has the form

$$V = \sum_i c_i x_i = \vec{c} \cdot \vec{x} \tag{3.65}$$

Geometrically, this linear constraint implies that the system's trajectories are strictly confined to a specific hyperplane in phase space.

Now, let us apply the Federer-Laurance formula to this type of constraints. First, note that we have

$$\nabla V = \vec{c} \to n^V = \hat{c} \tag{3.66}$$

To simplify the notation, for any vector $\vec{g}$ define

$$\vec{g}_\perp = \vec{g} - (\vec{g} \cdot \hat{c})\hat{c} \tag{3.67}$$

Now, the vectors ξ and n^ξ in the Federer-Laurance formula can be found as

$$\xi = \frac{\nabla H_\perp}{||\nabla H||} \Rightarrow n^\xi = \frac{\nabla H_\perp}{||\nabla H_\perp||} \tag{3.68}$$

and we can express the factor W as

$$W = \sqrt{\sum_{i<j} (\partial_i H c_j - c_i \partial_j H)^2} \tag{3.69}$$

By applying Lagrange's identity, the sum of squares of these 2×2 determinants can be rewritten as the product of the squared norms of the two vectors minus their squared dot product, *i.e.* we have

$$\sum_{i<j} (\partial_i H c_j - c_i \partial_j H)^2 = \left(\sum_i (\partial_i H)^2 \right) \left(\sum_i c_i^2 \right) - \left(\sum_i \partial_i H c_i \right)^2 \tag{3.70}$$

By using these formulae, at the end we get

$$\Phi(x) = \nabla \cdot \left(\frac{\nabla H_\perp}{||\nabla H_\perp||^2} \right) \tag{3.71}$$

Hence, we get a very similar formula to the Rugh formula.

We can generalize this formula (3.71) to systems with 3 conserved quantities, namely the Hamiltonian H and two linear constraints $V_1 = \vec{c}_1 \cdot \vec{x}$ and $V_2 = \vec{c}_2 \cdot \vec{x}$. In this case we have

$$\Phi(x) = \nabla \cdot \left(\frac{\nabla H_\perp}{||\nabla H_\perp||^2} \right) \tag{3.72}$$

where we define

$$\nabla H_\perp = \nabla H - (\nabla H \cdot \hat{c}_1)\hat{c}_1 - (\nabla H \cdot \hat{c}_2)\hat{c}_2 \tag{3.73}$$

3.5.3 Revisiting Mean Field ϕ^4 Theory, Again

Let us again consider the Hamiltonian

$$H_0 = \sum_i^N \left(\frac{p_i^2}{2} - \frac{1}{4}q_i^2 + \frac{1}{4}q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^N q_i q_j \tag{3.74}$$

and try to fix the magnetization

$$m = \frac{1}{N} \sum_{i=1}^N q_i \tag{3.75}$$

For this, we can use the Lagrange multiplier method and form the Hamiltonian

$$H = \sum_i^N \left(\frac{p_i^2}{2} - \frac{1}{4}q_i^2 + \frac{1}{4}q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^N q_i q_j + \lambda \left(\frac{1}{N} \sum_{i=1}^N q_i - m \right) \tag{3.76}$$

To solve λ, we consider the consistency condition

$$\frac{d}{dt} \left(\frac{1}{N} \sum_{i=1}^N q_i \right) = 0 \Rightarrow \sum_{i=1}^N p_i = 0 \tag{3.77}$$

As the second consistency condition, we have

$$\frac{d}{dt} \left(\sum_{i=1}^N p_i \right) = 0 \tag{3.78}$$

implying

$$\lambda = \sum_{i=1}^N (q_i - q_i^3) \tag{3.79}$$

Inserting this to (3.76) gives

$$H = \sum_i^N \left(\frac{p_i^2}{2} - \frac{1}{4}q_i^2 + \frac{1}{4}q_i^4 \right) - \frac{1}{4N} \sum_{i,j}^N q_i q_j + \left(\sum_{i=1}^N (q_i - q_i^3) \right) \left(\frac{1}{N} \sum_{i=1}^N q_i - m \right) \tag{3.80}$$

Hence, at the end we obtain a system with conserved Hamiltonian H and two linear constraints

$$\frac{1}{N} \sum_{i=1}^N q_i = m \qquad \sum_{i=1}^N p_i = 0 \tag{3.81}$$

Now, we can use the formalism in the previous section to calculate the temperature of this system, as we have done in Appendix E. It gives a result similar to those in Section 3.4.

Chapter 4

Statistical Mechanics of Systems with Modified Kinetic Energy

In this chapter, we will study the systems having the kinetic energy

$$K_p = 1 - \cos p \tag{4.1}$$

We changed the usual $p^2/2$ type kinetic energy to $1 - \cos p$ type kinetic energy, hence we can restrict the interval of p to $(-\pi, \pi)$. Notice that for low p we have

$$1 - \cos p \approx p^2/2 \tag{4.2}$$

hence we recover the usual kinetic energy.

The main advantage of the choice of $(1 - \cos p)$ instead of $p^2/2$ is that the former one gives a bounded kinetic energy while the latter gives a divergent kinetic energy in the limit $p \to \infty$.

4.1 Mean Field ϕ^4 Model

Let us consider the Hamiltonian

$$H = \sum_{i=1}^{N}(1 - \cos p_i) + \sum_{i=1}^{N} V(q_i) - \frac{1}{4N}\left(\sum_{i=1}^{N} q_i\right)^2 \tag{4.3}$$

where $V(q)$ is a local potential. We will focus on the case $V(q) = (-q^2 + q^4)/4$. In canonical ensemble, the partition function for our system is given as

$$Z = Z_p Z_q \tag{4.4}$$

where

$$Z_p = \left(\int_{-\pi}^{\pi} dp\, e^{-\beta(1-\cos p)} \right)^N \tag{4.5}$$

$$Z_q = \int dq_1 dq_2 \dots dq_N \exp\left[-\beta \left(\sum_{i=1}^N V(q_i) - \frac{1}{4N} \left(\sum_{i=1}^N q_i \right)^2 \right) \right] \tag{4.6}$$

Now, we can easily calculate Z_p in terms of modified Bessel function I_0:

$$Z_p \sim e^{-N\beta} I_0(\beta)^N = e^{-N(\beta - \ln I_0(\beta))} \tag{4.7}$$

However, we cannot easily obtain Z_q like this since there is a coupling term $(\sum_i q_i)^2$. To get away from this term, we will use the Hubbard-Stratonovich transformation

$$\exp(ba^2) = \frac{1}{\sqrt{4\pi b}} \int_{-\infty}^{\infty} dx\, \exp(-x^2/4b + xa) \tag{4.8}$$

Using this formula (4.8), we can simplify the expression for Z_q as

$$Z_q = \sqrt{\frac{N\beta}{\pi}} \int_{-\infty}^{\infty} dx\, e^{-N\beta x^2} \left(\int dq\, \exp -\beta(V(q) - xq) \right)^N \sim \int dx\, e^{NS(x,\beta)} \tag{4.9}$$

where we define $S(x, \beta)$ as

$$S(x, \beta) = -\beta x^2 + \ln \left(\int dq\, \exp -\beta(V(q) - xq) \right) \tag{4.10}$$

Now, in the large N limit, by Laplace method (4.9) reduces to

$$Z_q \sim e^{NS(x_0,\beta)} \tag{4.11}$$

where x_0 is the global maximum of $S(x, \beta)$, satisfying[1]

$$\frac{\partial S}{\partial x}(x_0, \beta) = 0 \tag{4.12}$$

The condition (4.12) gives

$$2x_0 = \frac{\int dq\, q\, e^{-\beta(V(q) - x_0 q)}}{\int dq\, e^{-\beta(V(q) - x_0 q)}} \equiv \langle q \rangle \tag{4.13}$$

Thus, physically $2x_0$ corresponds to the magnetization m. Hence, we can find a consistency relation for m as

[1] Strictly speaking, (4.12) is a condition for an extremum. It may be of interest to note that the condition for a maximum gives $\beta(\langle q^2 \rangle_{x_0} - \langle q \rangle_{x_0}^2) < 2$.

$$m = \frac{\int dq\, q\, e^{-\beta(V(q)-mq/2)}}{\int dq\, e^{-\beta(V(q)-mq/2)}} \tag{4.14}$$

Unfortunately, we cannot solve this equation (4.14) analytically, so we must use numerical methods. For given $V(q)$ and β, we can plot both sides of the equation by using the code presented in Appendix F.

We can also find m in terms of β by solving the consistency equation (4.14) with the code given in the Appendix G.

The result is as follows:

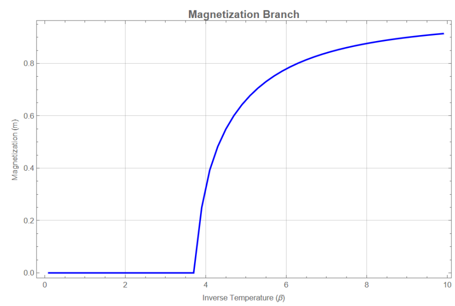

Figure 4.1: Magnetization vs. Inverse Temperature in Canonical Ensemble

As can be seen, a phase transition occurs at $\beta = \beta_c$. When $\beta < \beta_c$, i.e. $T > T_c$ we have a vanishing magnetization while for $\beta > \beta_c$, i.e. $T < T_c$ we have nonzero magnetization.

Now, we have Z_p and Z_q in our hands, we can calculate the total partition function Z and then the energy per particle $\epsilon = \overline{E}/N$:

$$\epsilon = 1 - \frac{I_1(\beta)}{I_0(\beta)} + \langle V(q) \rangle - \frac{m^2}{4}$$

In Appendix H, we give a code that plots ϵ and m as a function of $T = 1/\beta$. Here is the result:

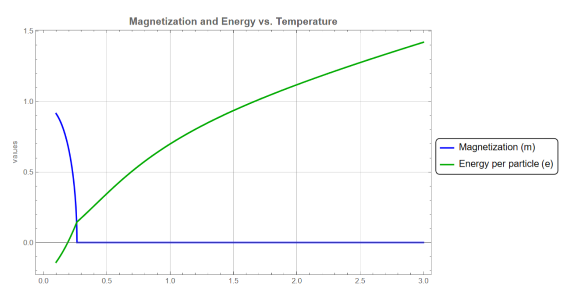

Figure 4.2: Magnetization and Energy vs. Temperature in Canonical Ensemble

As can be seen, at some finite temperature T_c, there happens a phase transition. For $T < T_c$ we have a nonvanishing magnetization and constant heat capacity while for $T > T_c$ we have zero magnetization and decreasing heat capacity.

We can also obtain a similar result in the Rugh formalism by using the program presented in Appendix I. Here is the result:

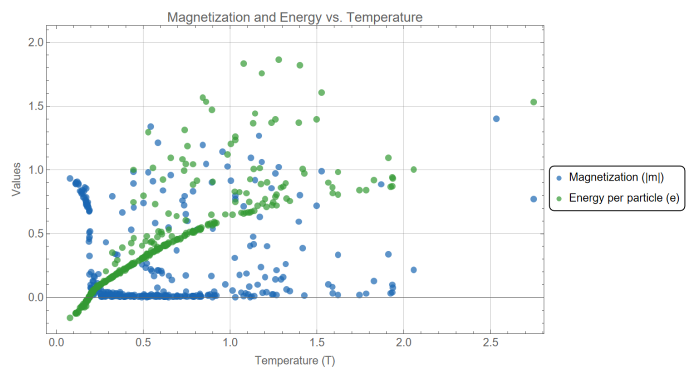

Figure 4.3: Magnetization and Energy vs. Temperature in Microcanonical Ensemble

At high energies, the phase diagram exhibits significant dispersion, with many states deviating from the expected thermodynamic curve, signaling us to the break of ergodicity. This is the direct consequence of dynamical localization caused by the Hamiltonian's kinetic speed limit. Because the kinetic term is strictly bounded to a maximum of 2.0 per particle, particles initialized with immense potential energy lack the kinetic capacity to traverse the macroscopic energy barriers.[2] As a result, specific particles remain permanently trapped outside the potential wells, generating a persistent, nonzero macroscopic magnetization.

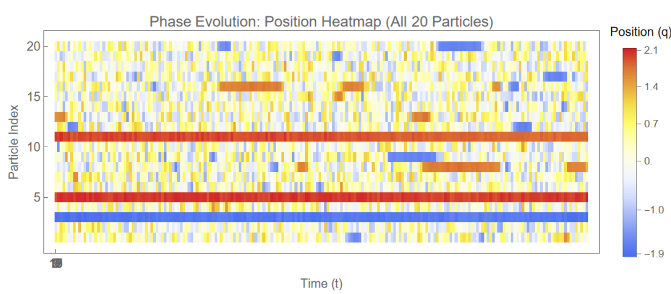

Figure 4.4: Time evolution of the $N = 20$ particle system. Notably, particles 3, 5, and 11 remain dynamically trapped outside the primary potential wells.

[2]While inter-particle interactions theoretically allow for energy exchange, the Kolmogorov-Arnold-Moser (KAM) theorem ensures that the invariant tori corresponding to these localized states are largely preserved.

4.2 XY Model in One Dimension

In this section, we will consider the Hamiltonian

$$H = \sum_{i=1}^{N}(1 - \cos p_i) + \sum_{i=1}^{N}(1 - \cos(q_{i+1} - q_i)) \tag{4.15}$$

where the canonical coordinates q_i and p_i are angular coordinates, and we have $q_{N+1} \equiv q_1$. Now, the kinetic part of the partition function is the same as the previous model:

$$Z_p \sim e^{-N(\beta - \ln I_0(\beta))} \tag{4.16}$$

and the position part of the partition function, as calculated in [20], is given as

$$Z_q \sim I_0(\beta)^N e^{-\beta N} \tag{4.17}$$

Using (4.16) and (4.17), one can calculate $\epsilon(T) = E/N$ as

$$\epsilon(T) = 2\left(1 - \frac{I_1(1/T)}{I_0(1/T)}\right) \tag{4.18}$$

Using this result, we can plot the temperature as a function of energy per particle, as illustrated in the graph below. Because the Hamiltonian consists exclusively of bounded trigonometric functions, the phase space is compact. Consequently, the caloric curve $T(\epsilon)$ lacks a high-energy linear regime; instead, the system reaches infinite temperature at the midpoint of its energy spectrum ($\epsilon = 2$), implying that any energy injection beyond this point will push the system into a regime of negative absolute temperature.

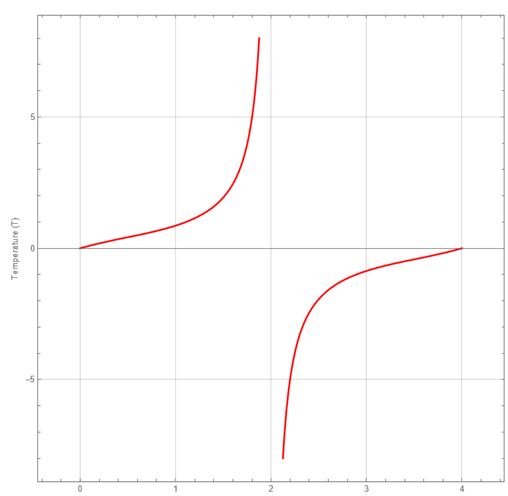

Figure 4.5: The Caloric Curve in Canonical Ensemble

We can also get a similar result by using Rugh formalism, via the first program that is given in Appendix J. The result is

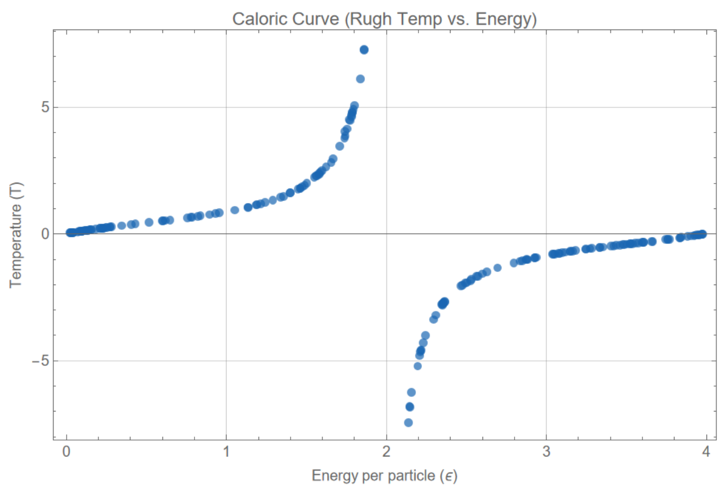

Figure 4.6: The Caloric Curve in Microcanonical Ensemble

As we can see, we have negative temperature for $\epsilon > 2$ and positive temperature for $\epsilon < 2$. The phase transition $\epsilon = 2$ corresponds to the most random configuration of the system, where all 'spins' are random.

The same program also gives the magnetic alignment

$$C(1) = \frac{1}{N} \sum_i \cos\left(q_{i+1} - q_i\right) \tag{4.19}$$

as a function of ϵ. Here is the result:

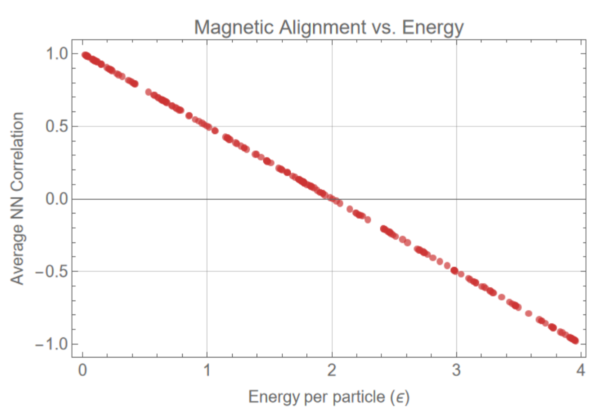

Figure 4.7: Magnetic Alignment vs. Energy

As can be seen, for low ϵ the spins are aligned, *i.e.* the system is in a ferromagnetic state while for high ϵ the spins are anti-aligned, *i.e.* the system is in an anti-ferromagnetic state.

The linearity of this graph can be explained as follows. Since in the Hamiltonian the position and momentum parts has similar forms, we have

$$N\epsilon = \langle H \rangle = \left\langle \sum_{i=1}^{N} (1 - \cos p_i) + \sum_{i=1}^{N} (1 - \cos (q_{i+1} - q_i)) \right\rangle = 2 \left\langle \sum_{i=1}^{N} (1 - \cos (q_{i+1} - q_i)) \right\rangle \tag{4.20}$$

implying

$$C(1) = 1 - \epsilon/2 \tag{4.21}$$

Now, similarly we can consider the correlation function

$$C(r) = \frac{1}{N} \sum_{i} \cos (q_{i+r} - q_i) \tag{4.22}$$

By numerical simulations, the correlation function has the following properties:

- For $\epsilon < 2$, $C(r)$ is a monotonic function of r and the spins are aligned, signaling ferromagnetism.
- For $\epsilon = 2$, $C(r) = 0$ hence we are in a completely random situation.
- For $\epsilon > 2$, $C(r)$ alternate signs when r changes and the spins are anti-aligned, signaling anti-ferromagnetism.

Chapter 5

Conclusions

In this report, we have systematically explored the intersection of Hamiltonian dynamics and equilibrium statistical mechanics, utilizing both analytical derivations and numerical simulations to highlight the deep geometric nature of thermodynamics.

By invoking the Birkhoff theorem and the ergodic hypothesis, we demonstrated that macroscopic thermodynamic observables can be accurately extracted directly from the geometric properties of the microcanonical phase space. The numerical implementation of the Rugh temperature formula, facilitated by robust symplectic integrators such as the Verlet algorithm, showed excellent agreement with theoretical predictions. For the mean-field ϕ^4 model, this dynamical approach successfully captured the continuous phase transition of the system, illustrating how particles escape local potential wells at high energies to yield a linear energy-temperature dependence consistent with the generalized equipartition theorem.

Furthermore, we successfully extended this geometric framework to systems possessing multiple conserved quantities, specifically by fixing the macroscopic magnetization. By employing Verlet algorithm, we maintained strict conservation laws without numerical drift. This allowed us to accurately map the constrained thermodynamics of the system, revealing how the heat capacity directly scales with the order parameter.

A particularly striking result of this study emerged in Chapter 4, where the assumption of a standard quadratic kinetic energy was relaxed in favor of a bounded, periodic kinetic energy. Through the application of the Hubbard-Stratonovich transformation and Laplace methodologies, we mapped the modified phase diagram of the mean-field ϕ^4 model. Also, in this model we observe the phenomenon of ergodicity breaking. Moreover, when applied to the 1D XY model, the bounded kinetic and potential energies forced the system's entropy to reach a maximum at a finite energy. Beyond this critical energy, the entropy decreases, directly resulting in a regime of negative absolute temperature.

In this exotic, highly energetic state, we showed that the system shifts from a ferromagnetically aligned state to a highly ordered, anti-aligned configuration.

Ultimately, this work reinforces the idea that thermodynamics is fundamentally encoded in the geometry of phase space. The ability to define temperature and entropy through phase-space volumes and vector field divergences not only clarifies the relation between the foundations of statistical mechanics and classical mechanics but also provides highly effective computational tools for investigating complex, constrained, and non-standard many-body systems.

Appendix A

In the following code, 'Nparts' stands for the number of particles in the system and 'alphas' stands for the various choices for α. The code first solves the equations of motions for the Hamiltonian (3.29), then calculates the average of Φ in the Rugh formalism to get the Rugh temperature. In the end, the code presents a comparison between our theoretical result and the numerical result.

```
ClearAll["Global`*"];

Nparts = 30;
alphas = {0.1, 0.09, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02, 0.01};

Print["Simulating Exact Dynamics... Please wait."];
results = {};

For[idx = 1, idx <= Length[alphas], idx++, alpha = alphas[[idx]];
  H = Sum[
     p[i]^2/2 - 1/4*q[i]^2 + 1/4*q[i]^4, {i, 1,
      Nparts}] - (alpha/(4*Nparts))*
     Sum[q[i]*q[j], {i, 1, Nparts}, {j, 1, Nparts}];
  vars = Flatten[Table[{q[i], p[i]}, {i, 1, Nparts}]];
  gradH = Table[D[H, v], {v, vars}];
  hessH = Table[D[H, v1, v2], {v1, vars}, {v2, vars}];
  gradH2 = gradH . gradH;
  lapH = Tr[hessH];
  PhiExact = lapH/gradH2 - 2*(gradH . hessH . gradH)/(gradH2^2);
  ruleTime =
   Flatten[Table[{q[i] -> q[i][t], p[i] -> p[i][t]}, {i, 1,
      Nparts}]];
  eqs = Flatten[
    Table[{q[i]'[t] == (D[H, p[i]] /. ruleTime),
      p[i]'[t] == (-D[H, q[i]] /. ruleTime)}, {i, 1, Nparts}]];
  initConds =
   Flatten[Table[{q[i][0] == 1/Sqrt[2.], p[i][0] == 0.}, {i, 1,
      Nparts}]];
  PhiDyn = PhiExact /. ruleTime;
```

```
  sys = Join[eqs,
    initConds, {intPhi'[t] == PhiDyn, intPhi[0] == 0.}];
  sysVars = Join[vars, {intPhi}];
  period = 2.0*Pi/Sqrt[1.0 - alpha/2.0];
  tMax = 5.0*period;
  sol = NDSolveValue[sys, sysVars, {t, 0, tMax}, MaxSteps -> Infinity];
  intPhiFunc = Last[sol];
  avgPhi = (intPhiFunc[tMax] - intPhiFunc[period])/(tMax - period);
  Tsimulated = 1/avgPhi;
  TfiniteN = (Nparts*alpha^2)/(8*(Nparts - 1)*Sqrt[4 - 2*alpha]);
  AppendTo[results, {alpha, Tsimulated, TfiniteN}];];

Grid[Prepend[
  results, {"\[Alpha]", "Simulated T (Exact Dynamics)",
   "T (Rugh Temperature)"}], Frame -> All,
 Background -> {None, {LightGray, None}}]
```

Appendix B

In this code, 'Npart' stands for the number of particles, 'dt' stands for the time intervals, 'steps' stands for the number of iterations done in a simulation, 'numSimulations' stands for the number of simulations done by the code and 'alpha' stands for α. The code first uses a symplectic integrator to solve numerically the equations of motions for given initial conditions determined randomly, then calculates the magnetization and the Rugh temperature. At the end, it plots both magnetization and Rugh temperature as a function of the energy.

```
ClearAll["Global`*"];

Npart = 300;
dt = 0.001;
steps = 30000;
numSimulations =
  300;
alpha = 0.1;


symplecticIntegratorCombined =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nSteps, _Integer}, {stepSize, _Real}},
     Module[{q = qIn, p = pIn, Np = Length[qIn], f, sumQAvg = 0.,
     invTsum = 0., p2},

       Do[

         f = 0.5 * q - q^3 + alpha * 0.5 * Total[q] / Np;

         p = p + 0.5 * stepSize * f;

         q = q + stepSize * p;

         f = 0.5 * q - q^3 + alpha * 0.5 * Total[q] / Np;

         p = p + 0.5 * stepSize * f;

         sumQAvg += Total[q];

         p2 = p · p;

         invTsum += (Np - 2.0) / (p2 + 10.^-10);

        , {i, nSteps}];

       {sumQAvg / nSteps, nSteps / invTsum}
      ]
    ];

calcEnergy[q_, p_] :=
  Total[p^2/2 - q^2/4 + q^4/4] - alpha * Total[q]^2 / (4 * Length[q]);

Print["Running Symplectic Simulations..."];
data = Table[
      Module[{q0, p0, E0, results, avgSumQ, T, scale, shift},

        scale = RandomReal[{0.2, 1.5}];
        shift = RandomChoice[{-1.0, 1.0}] * RandomReal[{0, 1.5}];

```

```
        q0 = RandomReal[{-scale, scale}, Npart] + shift;
        p0 = RandomReal[{-scale, scale}, Npart];

        E0 = calcEnergy[q0, p0];

        results = symplecticIntegratorCombined[q0, p0, steps, dt];
        avgSumQ = results[[1]];
        T = results[[2]];

        {E0, avgSumQ, T}
       ],
      {sim, numSimulations}
   ];

dataQ = Table[{pt[[1]], Abs[pt[[2]]]}, {pt, data}];

dataT = data[[All, {1, 3}]];

plotQ = ListPlot[dataQ,
     Frame -> True,
     FrameLabel -> {
        "Total Energy E",
        "|\!\(\*AngleBracketLeft\)\!\(\*UnderoverscriptBox[\(\[Sum]\),\
 \(i=1\), \(N\)]\)\!\(\*SubscriptBox[\(q\), \
\(i\)]\)\!\(\*AngleBracketRight\)|"
       },
     PlotLabel -> "Absolute Macroscopic Average vs. Energy",
     PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.8, 0.2, 0.2], Opacity[0.7]],
     ImageSize -> Medium,
     GridLines -> Automatic,
     BaseStyle -> {FontSize -> 12}
   ];

plotT = ListPlot[dataT,
     Frame -> True,
     FrameLabel -> {
        "Total Energy E",
        "Temperature T(E)"
       },
     PlotLabel -> "Caloric Curve (Rugh Temp vs. Energy)",
     PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7], Opacity[0.7]],
     ImageSize -> Medium,
     GridLines -> Automatic,
     BaseStyle -> {FontSize -> 12}
   ];

GraphicsRow[{plotQ, plotT}, ImageSize -> 1000]
```

Appendix C

In this code, 'Npart' stands for the number of particles, 'dt' stands for the time intervals, 'steps' stands for the number of iterations done in a simulation and 'alpha' stands for α. The code first uses a symplectic integrator to solve numerically the equations of motions for given initial conditions determined randomly, then calculates the total energy and the error in magnetization. At the end, it plots both the total energy and the numerical noise of the magnetization as a function of time.

```
    ClearAll["Global`*"];

Npart = 100;
dt = 0.001;
steps = 5000000;
alpha = 0.1;

q0Raw = RandomReal[{-3.0, 3.0}, Npart] + RandomReal[{0.0, 2.0}, Npart];
q0 = q0Raw;

p0Raw = RandomReal[{-1.0, 1.0}, Npart];
p0 = p0Raw - Mean[p0Raw];

integrateTrajectory =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nSteps, _Integer}, {stepSize, _Real}, {alphaVal, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn], f, Eval, mVal,
     history}, history = Table[{0., 0., 0.}, {nSteps + 1}];
    mVal = Total[q]/Np;
    f = 0.5*(q - Total[q]/Np) - q^3 + Total[q^3]/Np;
    Do[Eval =
      Total[p^2/2 - q^2/4 + q^4/4] - alphaVal*Total[q]^2/(4.0*Np);
     history[[i]] = {(i - 1)*stepSize, Eval, mVal};
     p = p + 0.5*stepSize*f;
     q = q + stepSize*p;
     mVal = Total[q]/Np;
     f = 0.5*(q - Total[q]/Np) - q^3 + Total[q^3]/Np;
     p = p + 0.5*stepSize*f;, {i, 1, nSteps}];
    Eval = Total[p^2/2 - q^2/4 + q^4/4] - alphaVal*Total[q]^2/(4.0*Np);
    mVal = Total[q]/Np;
    history[[nSteps + 1]] = {nSteps*stepSize, Eval, mVal};
    history]];

Print["Simulating trajectory..."];
trajData = integrateTrajectory[q0, p0, steps, dt, alpha];

energySeries = trajData[[All, {1, 2}]];

mInitial = trajData[[1, 3]];
magErrorSeries = Table[{pt[[1]], pt[[3]] - mInitial}, {pt, trajData}];

plotE = ListLinePlot[energySeries,
   PlotStyle -> Directive[RGBColor[0.8, 0.2, 0.2], Thickness[0.003]],
   Frame -> True, FrameLabel -> {"Time t", "Total Energy H(t)"},
```

```
    PlotLabel -> "Energy Conservation", GridLines -> Automatic,
    ImageSize -> Medium];

plotMError =
  ListLinePlot[magErrorSeries,
    PlotStyle -> Directive[RGBColor[0.1, 0.4, 0.7], Thickness[0.003]],
    Frame -> True,
    FrameLabel -> {"Time t", "Error \[Delta]m(t) = m(t) - m(0)"},
    PlotLabel -> "Numerical Noise", GridLines -> Automatic,
    ImageSize -> Medium, PlotRange -> All ];

GraphicsRow[{plotE, plotMError}, ImageSize -> 1000]
```

Appendix D

In this code, 'Npart' stands for the number of particles, 'mmag' stands for the fixed magnetization m, 'dt' stands for the time intervals, 'thermSteps' stands for the number of thermalization steps, 'steps' stands for the number of iterations done in a simulation and 'numSimulations' stands for the number of simulations done by the code. The code first uses a symplectic integrator to solve numerically the equations of motions derived via the Dirac-Bergmann formalism for given initial conditions determined randomly, then calculates the average order parameter and the Rugh temperature. At the end, it plots both the order parameter and the Rugh temperature as a function of the energy.

```
    ClearAll["Global`*"];

Npart = 300;
mmag = 0;
Mmag = Npart*mmag;
dt = 0.01;
thermSteps = 20000;
steps = 100000;
numSimulations = 600;

symplecticIntegratorCombined =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nTherm, _Integer}, {nSteps, _Integer}, {stepSize, _Real},
{M, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn] + 1, f, sumQAvg = 0.,
     invTsum = 0., p2, sumP, sumQ, twoKc}, sumQ = Total[q];
    f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
    Do[p = p + 0.5*stepSize*f;
     sumP = Total[p];
     q = q + stepSize*(p - sumP/Np);
     sumQ = Total[q];
     f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
     p = p + 0.5*stepSize*f;, {i, nTherm}];
    Do[p = p + 0.5*stepSize*f;
     sumP = Total[p];
     q = q + stepSize*(p - sumP/Np);
     sumQ = Total[q];
     f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
     p = p + 0.5*stepSize*f;
     sumQAvg += sumQ;
     sumP = Total[p];
     p2 = p . p;
     twoKc = p2 - sumP^2/Np;
     invTsum += (Np - 3.0)/(twoKc + 10.^-10);, {i, nSteps}];
    {sumQAvg/nSteps, nSteps/invTsum}]];

calcEnergy[q_, p_, M_] :=
  Module[{Np = Length[q] + 1, sumQ = Total[q],
```

```
    sumP = Total[p]}, (Total[p^2]/2 - sumP^2/(2*Np)) +
    Total[-q^2/4 + q^4/4] - (M - sumQ)^2/4 + (M - sumQ)^4/4 -
    M^2/(4*Np)];

Print["Running Symplectic Simulations..."];
data = Table[
   Module[{qTemp, q0, pTemp, p0, E0, results, avgSumQ, T, scale},
    scale = RandomReal[{0.2, 1.4}];
    qTemp = RandomReal[{-scale, scale}, Npart];
    qTemp = qTemp - (Total[qTemp] - Mmag)/Npart;
    q0 = qTemp[[1 ;; Npart - 1]];
    pTemp = RandomReal[{-scale, scale}, Npart];
    pTemp = pTemp - Mean[pTemp];
    p0 = pTemp[[1 ;; Npart - 1]] - pTemp[[-1]];
    E0 = calcEnergy[q0, p0, Mmag];
    results =
     symplecticIntegratorCombined[q0, p0, thermSteps, steps, dt, Mmag];
    avgSumQ = results[[1]]/(Npart - 1);
    T = results[[2]];
    {E0, avgSumQ, T}], {sim, numSimulations}];

dataQ = Table[{pt[[1]], Abs[pt[[2]]]}, {pt, data}];
dataT = data[[All, {1, 3}]];

plotQ = ListPlot[dataQ, Frame -> True,
   FrameLabel -> {"Total Energy E", "Average |Sum(Q)|"},
   PlotLabel -> "Order Parameter vs Energy",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.8, 0.2, 0.2],
     Opacity[0.7]], ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

plotT = ListPlot[dataT, Frame -> True,
   FrameLabel -> {"Total Energy E", "Temperature T(E)"},
   PlotLabel -> "Caloric Curve (Rugh Temp vs. Energy)",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7],
     Opacity[0.7]], ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

GraphicsRow[{plotQ, plotT}, ImageSize -> 1000]
```

In this other code, 'Npart' stands for the number of particles, 'mValues' stands for the various choices for the fixed magnetization m, 'dt' stands for the time intervals, 'thermSteps' stands for the number of thermalization steps, 'steps' stands for the number of iterations done in a simulation and 'numSimulations' stands for the number of simulations done by the code. The code first uses a symplectic integrator to solve numerically the equations of motions derived via the Dirac-Bergmann formalism for given initial conditions determined randomly, then calculates the total energy and the Rugh temperature for each choice of m. At the end, it plots the Rugh temperature as a function of the energy to visually demonstrate the superposed caloric curves.

```
    ClearAll["Global`"];
Npart = 300;
dt = 0.01;
thermSteps = 20000;
steps = 100000;
numSimulations = 300;
mValues = {0.0, 0.2, 0.4, 0.6, 0.8, 1.0};

symplecticIntegratorCombined =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nTherm, _Integer}, {nSteps, _Integer}, {stepSize, _Real}, \
{M, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn] + 1.0, f, sumQAvg = 0.,
     invTsum = 0., p2, sumP, sumQ, twoKc}, sumQ = Total[q];
```

```
    f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
    sumP = Total[p];
    Do[p = p + 0.5*stepSize*f;
     q = q + stepSize*(p - sumP/Np);
     sumQ = Total[q];
     f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
     p = p + 0.5*stepSize*f;, {i, nTherm}];
    Do[p = p + 0.5*stepSize*f;
     sumP = Total[p];
     q = q + stepSize*(p - sumP/Np);
     sumQ = Total[q];
     f = 0.5*q - q^3 - 0.5*(M - sumQ) + (M - sumQ)^3;
     p = p + 0.5*stepSize*f;
     sumQAvg += sumQ;
     sumP = Total[p];
     p2 = p . p;
     twoKc = p2 - sumP^2/Np;
     invTsum += (Np - 3.0)/(twoKc + 10.^-10);, {i, nSteps}];
    {sumQAvg/nSteps, nSteps/invTsum}]];

calcEnergy[q_, p_, M_] :=
  Module[{Np = Length[q] + 1, sumQ = Total[q],
    sumP = Total[p]}, (Total[p^2]/2 - sumP^2/(2  Np)) +
    Total[-q^2/4 + q^4/4] - (M - sumQ)^2/4 + (M - sumQ)^4/4 -
    M^2/(4  Np)];

Print["Running Simulations..."];
allDataT = {};

For[idx = 1, idx <= Length[mValues], idx++, mmag = mValues[[idx]];
  Mmag = Npart*mmag;
  dataT =
   Table[Module[{scale, shift, qTemp, q0, pTemp, p0, E0, results, T},
     scale = RandomReal[{0.2, 1.4}];
     qTemp = RandomReal[{-scale, scale}, Npart];
     qTemp = qTemp - (Total[qTemp] - Mmag)/Npart;
     q0 = qTemp[[1 ;; Npart - 1]];
     pTemp = RandomReal[{-scale, scale}, Npart];
     pTemp = pTemp - Mean[pTemp];
     p0 = pTemp[[1 ;; Npart - 1]];
     E0 = calcEnergy[q0, p0, Mmag];
     results =
      symplecticIntegratorCombined[q0, p0, thermSteps, steps, dt,
       Mmag];
     T = results[[2]];
     {E0, T}], {sim, numSimulations}];
  AppendTo[allDataT, dataT];];

plotT = ListPlot[allDataT, Frame -> True,
   FrameLabel -> {"Total Energy E", "Temperature T(E)"},
   PlotLabel -> "Superposed Caloric Curves (Rugh Temp vs. Energy)",
   PlotLegends ->
    Placed[PointLegend[("m = " <> ToString[#] & /@ mValues),
      LegendFunction -> "Frame"], Right],
   PlotStyle -> {Directive[PointSize[0.012], RGBColor[0.8, 0.2, 0.2],
      Opacity[0.7]],
     Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7],
      Opacity[0.7]],
     Directive[PointSize[0.012], RGBColor[0.2, 0.6, 0.2],
      Opacity[0.7]]}, ImageSize -> Large, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

Print[plotT];
```

Appendix E

In this code, 'Npart' stands for the number of particles, 'Mmag' stands for the fixed magnetization m, 'dt' stands for the time intervals, 'thermSteps' stands for the number of thermalization steps, 'steps' stands for the number of iterations done in a simulation and 'numSimulations' stands for the number of simulations done by the code. The code first uses a symplectic integrator to solve numerically the equations of motions for a system subject to two linear constraints for given initial conditions determined randomly, then calculates the average magnetization and the geometric Rugh temperature. At the end, it plots both the constraint verification and the geometric Rugh temperature as a function of the energy.

```
ClearAll["Global`*"];
Npart = 300;
Mmag = 1.0;
dt = 0.01;
thermSteps = 20000;
steps = 100000;
numSimulations = 300;
symplecticIntegratorCombined = Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nTherm, _Integer}, {nSteps, _Integer}, {stepSize, _Real},
{mTarget, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn]*1.0, f, Gq, Gp, G2,
     divG, GTdGG, Phi, invTsum = 0.0}, Do[f = 0.5*q - q^3;
     f = f - Total[f]/Np; p = p + 0.5*stepSize*f;
     q = q + stepSize*p;
     f = 0.5*q - q^3;
     f = f - Total[f]/Np;
     p = p + 0.5*stepSize*f;, {i, nTherm}];
    Do[f = 0.5*q - q^3;
     f = f - Total[f]/Np;
     p = p + 0.5*stepSize*f;
     q = q + stepSize*p;
     f = 0.5*q - q^3;
     f = f - Total[f]/Np;
     p = p + 0.5*stepSize*f;
     q = q - Total[q]/Np + mTarget;
     p = p - Total[p]/Np;
     Gq = -f;
     Gp = p;
     G2 = Total[Gq^2] + Total[Gp^2] + 10.^-10;
     divG = 3.0*(1.0 - 1.0/Np)*Total[q^2] - Np/2.0 + Np - 0.5;
     GTdGG = Total[Gq^2*(3.0*q^2 - 0.5)] + Total[Gp^2];
     Phi = divG/G2 - 2.0*GTdGG/(G2^2);
     invTsum += Phi;, {i, nSteps}];
    {Total[q]/Np, nSteps/invTsum}]];
calcEnergy[q_, p_, mTarget_] := Total[p^2/2 - q^2/4 + q^4/4] - Length[q]*mTarget^2/4.0;
```

```
Print["Running Constrained Geometric Symplectic Simulations..."];
data = Table[
   Module[{scale, shift, qTemp, q0, pTemp, p0, E0, results, T,
     finalM}, scale = RandomReal[{0.2, 1.5}];
    shift = RandomChoice[{-1.0, 1.0}]*RandomReal[{0, 1.5}];
    qTemp = RandomReal[{-scale, scale}, Npart] + shift;
    q0 = qTemp - Total[qTemp]/Npart + Mmag;
    pTemp = RandomReal[{-scale, scale}, Npart];
    p0 = pTemp - Total[pTemp]/Npart;
    E0 = calcEnergy[q0, p0, Mmag];
    results =
     symplecticIntegratorCombined[q0, p0, thermSteps, steps, dt, Mmag];
    finalM = results[[1]];
    T = results[[2]];
    {E0, finalM, T}], {sim, numSimulations}];
dataT = data[[All, {1, 3}]];
dataQ = data[[All, {1, 2}]];
plotT = ListPlot[dataT, Frame -> True,
   FrameLabel -> {"Total Energy E", "Temperature T(E)"},
   PlotLabel -> "Caloric Curve (Geometric Rugh vs. Energy)",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7],
     Opacity[0.7]], ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];
plotQ = ListPlot[dataQ, Frame -> True,
   FrameLabel -> {"Total Energy E", "Average Sum(Q)/N"},
   PlotLabel ->
    "Constraint Verification (Fixed m=" <> ToString[Mmag] <> ")",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.8, 0.2, 0.2],
     Opacity[0.7]], PlotRange -> {Mmag - 0.05, Mmag + 0.05},
   ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];
GraphicsRow[{plotQ, plotT}, ImageSize -> 1000]
```

Appendix F

In the following code, 'V[q]' stands for the local potential and 'beta' stands for the inverse temperature β. The code first calculates the right-hand side of the consistency equation using numerical integration. In the end, it plots both the left-hand side and the right-hand side of the consistency equation as a function of the magnetization to visually demonstrate their intersections.

```
V[q_] := (q^4 - q^2)/4;

beta = 2.0;

RHS[m_?NumericQ] :=
  NIntegrate[q*Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}]/
   NIntegrate[Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}];

Plot[{m, RHS[m]}, {m, -3, 3},
 PlotStyle -> {Directive[Thick, Blue], Directive[Thick, Dashed, Red]},
  PlotLegends -> {"LHS = m", "RHS"},
 AxesLabel -> {"Magnetization (m)", "Value"},
 PlotLabel -> Style["Consistency Equation: m vs RHS", 14, Bold],
 GridLines -> Automatic, ImageSize -> Large]
```

Appendix G

In the following code, 'V[q]' stands for the local potential and 'betaPoints' stands for the various choices for the inverse temperature β. The code first uses a numerical root-finding algorithm to solve the consistency equation to find the exact magnetization for a given β. In the end, it plots the magnetization branch as a function of the inverse temperature.

```
V[q_] := (-q^2 + q^4)/4;

RHS[m_?NumericQ, beta_?NumericQ] :=
  NIntegrate[q*Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}]/
   NIntegrate[Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}];

FindM[beta_?NumericQ] :=
  Module[{sol, mVal},
   Quiet[sol = FindRoot[m == RHS[m, beta], {m, 1.0}];];
   mVal = m /. sol;
   If[mVal < 10^-5, 0, mVal]];
betaPoints = Range[0.1, 10, 0.2];
mPoints = Table[{b, FindM[b]}, {b, betaPoints}];

ListLinePlot[mPoints, Frame -> True,
 FrameLabel -> {"Inverse Temperature (\[Beta])", "Magnetization (m)"},
  PlotLabel -> Style["Magnetization Branch", 14, Bold],
 PlotStyle -> Directive[Thick, Blue], GridLines -> Automatic,
 ImageSize -> Large]
```

Appendix H

In the following code, 'V[q]' stands for the local potential and 'tPoints' stands for the various choices for the temperature T. The code first uses a numerical root-finding algorithm to find the magnetization for a given temperature, then calculates the expected potential energy and the modified kinetic energy to get the total energy per particle. In the end, it plots both the magnetization and the energy per particle as a function of the temperature.

```
V[q_] := (-q^2 + q^4)/4;

RHS[m_?NumericQ, beta_?NumericQ] :=
  NIntegrate[q*Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}]/
   NIntegrate[Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}];

FindM[beta_?NumericQ] :=
  Module[{sol, mVal},
   Quiet[sol = FindRoot[m == RHS[m, beta], {m, 1.0}];];
   mVal = m /. sol;
   If[mVal < 10^-5, 0, mVal]];

ExpV[m_?NumericQ, beta_?NumericQ] :=
  NIntegrate[
    V[q]*Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}]/
   NIntegrate[Exp[-beta*(V[q] - m*q/2)], {q, -Infinity, Infinity}];

EnergyPerParticleT[T_?NumericQ] :=
  Module[{beta, mVal, vAvg, kinetic, eTotal}, beta = 1.0/T;
   mVal = FindM[beta];
   vAvg = ExpV[mVal, beta];
   kinetic = 1 - BesselI[1, beta]/BesselI[0, beta];
   eTotal = kinetic + vAvg - (mVal^2)/4;
   eTotal];

tPoints = Range[0.1, 5.0, 0.01];
Print["Calculating, please wait..."];
ePointsT = Table[{T, EnergyPerParticleT[T]}, {T, tPoints}];

ListLinePlot[ePointsT, Frame -> True,
 FrameLabel -> {"Temperature (T)", "Energy per particle (e)"},
 PlotLabel -> Style["Average Energy vs. Temperature", 14, Bold],
 PlotStyle -> Directive[Thick, Darker[Green]], GridLines -> Automatic,
  ImageSize -> Large]
```

Appendix I

In this code, 'Npart' stands for the number of particles, 'dt' stands for the time intervals, 'thermSteps' stands for the number of thermalization steps, 'steps' stands for the number of iterations done in a simulation and 'numSimulations' stands for the number of simulations done by the code. The code first uses a symplectic integrator to solve numerically the equations of motions for the system with modified kinetic energy for given initial conditions determined randomly, then calculates the magnetization, the energy per particle and the Rugh temperature. At the end, it plots both the magnetization and the energy per particle as a function of the temperature.

```
ClearAll["Global`*"];
Npart = 400;
dt = 0.01;
thermSteps = 8000000;
steps = 10000000;
numSimulations = 300;

symplecticIntegratorCombined =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nTherm, _Integer}, {nSteps, _Integer}, {stepSize, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn]*1.0, f, sumQAvg = 0.,
     invTsum = 0., sumE = 0., mVal, sinp, cosp, sin2p, sumCos,
     sumSin2, sumSin2Cos}, mVal = Total[q]/Np;
    f = 0.5*q - q^3 + 0.5*mVal;
    Do[p = p + 0.5*stepSize*f;
     q = q + stepSize*Sin[p];
     mVal = Total[q]/Np;
     f = 0.5*q - q^3 + 0.5*mVal;
     p = p + 0.5*stepSize*f;, {i, nTherm}];
    Do[p = p + 0.5*stepSize*f;
     sinp = Sin[p];
     cosp = Cos[p];
     q = q + stepSize*sinp;
     mVal = Total[q]/Np;
     f = 0.5*q - q^3 + 0.5*mVal;
     p = p + 0.5*stepSize*f;
     sumQAvg += mVal;
     sin2p = sinp*sinp;
     sumCos = Total[cosp];
     sumSin2 = Total[sin2p];
     sumSin2Cos = sin2p . cosp;
     sumE +=
      Np - sumCos + Total[-0.25*q^2 + 0.25*q^4] - 0.25*Np*mVal^2;
     invTsum +=
      sumCos/(sumSin2 + 10.^-10) -
       2.0*sumSin2Cos/(sumSin2^2 + 10.^-10);, {i, nSteps}];
    {Abs[sumQAvg/nSteps], sumE/(nSteps*Np), nSteps/invTsum}],
   RuntimeOptions -> "Speed"];
```

```

Print["Running Fast Symplectic Simulations in Parallel... Please \
wait."];

DistributeDefinitions[symplecticIntegratorCombined, Npart, thermSteps,
   steps, dt];

data = ParallelTable[
   Module[{q0, p0, results, scaleQ, scaleP, shift},
    scaleQ = RandomReal[{0.2, 1.5}];
    scaleP = RandomReal[{0.2, 2.5}];
    shift = RandomChoice[{-1.0, 1.0}]*RandomReal[{0, 1.5}];
    q0 = RandomReal[{-scaleQ, scaleQ}, Npart] + shift;
    p0 = RandomReal[{-scaleP, scaleP}, Npart];
    results =
     symplecticIntegratorCombined[q0, p0, thermSteps, steps, dt];
    results], {sim, numSimulations}];

data = Select[data, #[[3]] > 0 &];
data = SortBy[data, #[[3]] &];

dataM = Table[{pt[[3]], pt[[1]]}, {pt, data}];
dataE = Table[{pt[[3]], pt[[2]]}, {pt, data}];

plot = ListPlot[{dataM, dataE}, Frame -> True,
   FrameLabel -> {"Temperature (T)", "Values"},
   PlotLabel -> "Magnetization and Energy vs. Temperature",
   PlotLegends ->
    Placed[PointLegend[{"Magnetization (|m|)",
       "Energy per particle (e)"}, LegendFunction -> "Frame"], Right],
    PlotStyle -> {Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7],
       Opacity[0.7]],
     Directive[PointSize[0.012], RGBColor[0.2, 0.6, 0.2],
      Opacity[0.7]]}, ImageSize -> Large, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

Print[plot];
```

Appendix J

In this code, 'Npart' stands for the number of particles, 'dt' stands for the time intervals, 'thermSteps' stands for the number of thermalization steps, 'steps' stands for the number of iterations done in a simulation and 'numSimulations' stands for the number of simulations done by the code. The code first uses a symplectic integrator to solve numerically the equations of motions for the one-dimensional XY model for given initial conditions determined randomly, then calculates the energy per particle and the Rugh temperature. At the end, it plots the Rugh temperature as a function of the energy per particle.

```
ClearAll["Global`"];
Npart = 200;
dt = 0.01;
thermSteps = 10000;
steps = 40000;
numSimulations = 200;

symplecticIntegratorCombined =
  Compile[{{qIn, _Real, 1}, {pIn, _Real,
     1}, {nTherm, _Integer}, {nSteps, _Integer}, {stepSize, _Real}},
   Module[{q = qIn, p = pIn, Np = Length[qIn], f, sumE = 0.,
     invTsum = 0., sumSin2 = 0., sumCos = 0., sumSin2Cos = 0., qLeft,
     qRight, sumCosDiff = 0.}, Do[qRight = RotateLeft[q, 1];
     qLeft = RotateRight[q, 1];
     f = Sin[qRight - q] - Sin[q - qLeft];
     p = p + 0.5*stepSize*f;
     q = q + stepSize*Sin[p];
     qRight = RotateLeft[q, 1];
     qLeft = RotateRight[q, 1];
     f = Sin[qRight - q] - Sin[q - qLeft];
     p = p + 0.5*stepSize*f;, {i, nTherm}];
    Do[qRight = RotateLeft[q, 1];
     qLeft = RotateRight[q, 1];
     f = Sin[qRight - q] - Sin[q - qLeft];
     p = p + 0.5*stepSize*f;
     q = q + stepSize*Sin[p];
     qRight = RotateLeft[q, 1];
     qLeft = RotateRight[q, 1];
     f = Sin[qRight - q] - Sin[q - qLeft];
     p = p + 0.5*stepSize*f;
     sumE += Total[1.0 - Cos[p]] + Total[1.0 - Cos[qRight - q]];
     sumCosDiff += Total[Cos[qRight - q]];
     sumSin2 += Total[Sin[p]^2];
     sumCos += Total[Cos[p]];
     sumSin2Cos += Total[Sin[p]^2*Cos[p]];
     invTsum +=
      sumCos/(sumSin2 + 10.^-10) -
       2.0*sumSin2Cos/(sumSin2^2 + 10.^-10);, {i, nSteps}];
    {sumE/(nSteps*Np), nSteps/invTsum, sumCosDiff/(nSteps*Np)}]];
```

```
data = Table[
   Module[{q0, p0, results, type, k}, type = RandomChoice[{1, 2}];
    k = RandomReal[{0.2, 3.0}];
    If[type == 1, q0 = RandomReal[{-k, k}, Npart];
     p0 = RandomReal[{-k, k}, Npart];,
     q0 = Table[If[EvenQ[i], Pi, 0], {i, Npart}] +
       RandomReal[{-k, k}, Npart];
     p0 = RandomReal[{Pi - k, Pi + k}, Npart];];
    results =
     symplecticIntegratorCombined[q0, p0, thermSteps, steps, dt];
    results], {sim, numSimulations}];

dataT = data[[All, {1, 2}]];
dataC = data[[All, {1, 3}]];

plotT = ListPlot[dataT, Frame -> True,
   FrameLabel -> {"Energy per particle (\[Epsilon])",
     "Temperature (T)"},
   PlotLabel -> "Caloric Curve (Rugh Temp vs. Energy)",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.1, 0.4, 0.7],
     Opacity[0.7]], ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

plotC = ListPlot[dataC, Frame -> True,
   FrameLabel -> {"Energy per particle (\[Epsilon])",
     "Average NN Correlation"},
   PlotLabel -> "Magnetic Alignment vs. Energy",
   PlotStyle ->
    Directive[PointSize[0.012], RGBColor[0.8, 0.2, 0.2],
     Opacity[0.7]], ImageSize -> Medium, GridLines -> Automatic,
   BaseStyle -> {FontSize -> 12}];

GraphicsRow[{plotT, plotC}, ImageSize -> 1000]
```

Appendix K

A Brief Introduction to Ergodicity

K.1 Dynamical Systems

In this appendix, we will give a brief introduction to ergodicity. For a more detailed exposition, see [21]. Ergodic theory analyzes dynamical systems from a probabilistic point of view, hence let us first recover basic definitions of dynamical systems.

A dynamical system is a pair (M, Φ) such that

- M is a set called the phase space, where each point in M represents a possible configuration of the dynamical system under the consideration. Generally M has another structures defined on it, for instance M may be a topological space or manifold.
- $\Phi = \{\Phi^t : M \to M\}$ is a set of transformations which forms a one-parameter group, *i.e.* one have

$$\Phi^t \circ \Phi^s = \Phi^{t+s} \qquad (\Phi^t)^{-1} = \Phi^{-t} \qquad \Phi^0 = id_M$$

  In there, t takes values from either $\mathbb{Z}$ or $\mathbb{R}$. In the former case, the dynamical system is called discrete while in the latter case it is called continuous.

For our purposes, we will assume one more property, the existence of an invariant measure μ:

- The invariant measure on μ satisfies $\mu(M) = 1$ and for all t and measurable subset $A \subseteq M$

$$\mu(\Phi^t(A)) = \mu(A)$$

Let us give some examples to show the generality of the concept of dynamical systems.

Example (Flow on Torus): Let $v = (v_1, v_2) \in \mathbb{R}^2$ be a vector. Consider the dynamical system

- $M = T^2 = \mathbb{R}^2/\mathbb{Z}^2$
- $\Phi^t(x) = x + vt \pmod 1$, where $t \in \mathbb{R}$
- $\mu =$ Lebesgue measure

This defines a dynamical system, as can be checked easily. The behavior of this dynamical system heavily depends to the rationality or the irrationality of the number $\alpha = v_1/v_2$.

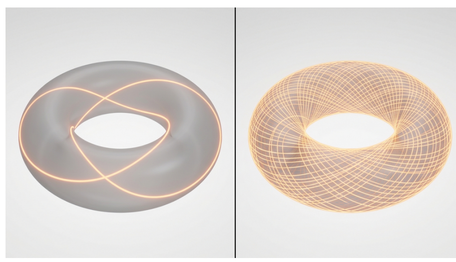

Figure K.1: In the left, you see the case $\alpha \in \mathbb{Q}$ while in the right you see the case $\alpha \notin \mathbb{Q}$.

Example: A classical mechanical system is an example to a dynamical system. More explicitly, consider the dynamical system defined as

- $M =$ Any symplectic manifold
- $\Phi^t(x) =$ Hamiltonian flow of a function $H \in C^\infty(M)$, where $t \in \mathbb{R}$
- $\mu =$ Liouville volume form

Example (Discrete translations on S^1): We can consider a simplified version of the first example as follows: $(\alpha \in \mathbb{R})$

- $M = S^1 =$ Unit circle
- $\Phi^t(x) = x + \alpha t \pmod{2\pi}$, where $t \in \mathbb{Z}$
- $\mu =$ Lebesgue measure

The dynamics of this system is again determined by whether α is rational or not. If $\alpha \in \mathbb{Q}$, the system is periodic while if $\alpha \notin \mathbb{Q}$ we have the following theorem:

Theorem K.1.1. *If $\alpha \notin \mathbb{Q}$, then the trajectories of any point $p_0 \in S^1$ are dense in S^1.*

Proof. Let $S^1 = \mathbb{R}/\mathbb{Z}$. We want to show that the forward orbit $\mathcal{O}(p_0) = \{p_0 + n\alpha \pmod 1 \mid n \in \mathbb{N}\}$ is dense in S^1. Because translation by p_0 is a continuous map that preserves distances, it suffices to prove this for the case $p_0 = 0$.

Let $N \in \mathbb{N}$ be an arbitrary integer, and divide $[0, 1)$ into N equal sub-intervals of length $\frac{1}{N}$. Consider the first $N + 1$ points in the orbit of 0:

$$0, \alpha, 2\alpha, \ldots, N\alpha \pmod 1$$

By the Pigeonhole Principle, at least two distinct points, say $i\alpha$ and $j\alpha$ (with $0 \le i < j \le N$), must fall into the same sub-interval.

Let $k = j - i$. The integer k satisfies $0 < k \le N$, and the distance between 0 and $k\alpha \pmod 1$ is strictly less than $\frac{1}{N}$. Because $\alpha \notin \mathbb{Q}$, $k\alpha$ cannot be an integer. Therefore, $k\alpha \equiv \epsilon \pmod 1$ for some non-zero ϵ satisfying $0 < |\epsilon| < \frac{1}{N}$.

Now consider the sub-sequence generated by taking steps of size k:

$$0, k\alpha, 2k\alpha, 3k\alpha, \ldots \pmod 1$$

This corresponds to taking successive steps of size ϵ around the circle. Since the step size is strictly less than $\frac{1}{N}$, these points form a grid on S^1 where the gap between adjacent points is less than $\frac{1}{N}$. Consequently, every point $y \in S^1$ lies within a distance of $\frac{1}{N}$ from some point in the orbit of 0.

Since N was chosen arbitrarily, we can make $\frac{1}{N}$ arbitrarily small, proving the orbit of 0 is dense in S^1. □

K.2 Time and Phase Averages

For a dynamical system, there is two kind of averages of a function $f : M \to \mathbb{R}$ we can consider:

- The time average, which is defined as

$$\langle f \rangle_t (x_0) = \begin{cases} \lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\sum_{t=0}^{T-1} f(\Phi^t(x_0)) & \text{for a discrete system} \\ \lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\int_0^T f(\Phi^t(x_0))\, dt & \text{for a continuous system} \end{cases}$$

  It is of course *a priori* not obvious that whether the limit in the definitions exists. In the future, we will show that $\langle f \rangle_t (x_0)$ exists for a.e. $x_0 \in M$. (The celebrated theorem of Birkhoff.)

- The phase space average, which is defined as

$$\langle f \rangle_p = \int_M f \, d\mu$$

  (Recall that $\mu(M) = 1$) The phase space average exists provided that $f \in L^1(M, \mu)$.

Let us state and prove the Birkhoff theorem for the discrete dynamical systems[1]:

Theorem K.2.1 (Birkhoff Theorem)**.** *Let (M, μ, Φ) be a discrete measure-preserving dynamical system, where $\mu(M) = 1$. For any $f \in L^1(M, \mu)$, the time average*

$$\langle f \rangle_t (x_0) = \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} f(\Phi^t(x_0))$$

exists for a.e. $x_0 \in M$.

Proof. Without proof, we will assume the following rather technical lemma:

Lemma K.2.1 (Maximal Ergodic Lemma)**.** *For any $g \in L^1(M, \mu)$, if we define the set*

$$M^+ = \left\{ x_0 \in M \;\middle|\; \sup_{T \geq 1} \sum_{t=0}^{T-1} g(\Phi^t(x_0)) > 0 \right\}$$

then we have

$$\int_{M^+} g \, d\mu \geq 0$$

We define the upper and lower bounds of the time averages:

$$f^*(x_0) = \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} f(\Phi^t(x_0)) \quad \text{and} \quad f_*(x_0) = \liminf_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} f(\Phi^t(x_0)).$$

Suppose $f_*(x_0) \neq f^*(x_0)$ on a set of positive measure. Then there exist rational numbers $\alpha < \beta$ such that the set

$$E_{\alpha,\beta} = \{x_0 \in M \mid f_*(x_0) < \alpha < \beta < f^*(x_0)\}$$

has $\mu(E_{\alpha,\beta}) > 0$. Because time shifts do not alter the infinite limit of the averages, $E_{\alpha,\beta}$ is strictly Φ-invariant.

Applying the Maximal Ergodic Lemma to the test function $f(x_0) - \beta$ restricted to $E_{\alpha,\beta}$, we know the integral over this set is non-negative, yielding:

$$\int_{E_{\alpha,\beta}} f \, d\mu \geq \beta \mu(E_{\alpha,\beta}).$$

Applying the same lemma to $\alpha - f(x_0)$ yields:

$$\int_{E_{\alpha,\beta}} f \, d\mu \leq \alpha \mu(E_{\alpha,\beta}).$$

Thus, $\beta\mu(E_{\alpha,\beta}) \leq \alpha\mu(E_{\alpha,\beta})$. Since $\mu(E_{\alpha,\beta}) > 0$, this implies $\beta \leq \alpha$, contradicting our choice of $\alpha < \beta$. Therefore, $\mu(E_{\alpha,\beta}) = 0$. Taking the countable union over all rational pairs $\alpha < \beta$ shows that $f^*(x_0) = f_*(x_0)$ almost everywhere. The limit $\langle f \rangle_t (x_0)$ exists.

□

[1] The proof for continuous dynamical systems is similar. From now on, we will prove all theorems for discrete systems, but the statements will hold for continuous systems too with similar proofs.

K.3 Ergodicity

Let (M, μ, Φ) be a discrete dynamical system. Then, the following are equivalent:

- **E1:** For any measurable $A \subseteq M$, the time spent in A by a generic trajectory is given by the measure of A:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \chi_A(\Phi^t(x_0)) = \mu(A) \quad \text{for a.e. } x_0 \in M$$

- **E2:** For any $f \in L^1(M, \mu)$, the time and phase averages coincides almost everywhere:
$$\langle f \rangle_t(x_0) = \langle f \rangle_p \quad \text{for a.e. } x_0 \in M$$

- **E3:** No constant of motions exists but the trivial ones, namely if $f \in L^1(M, \mu)$ we have

$$f(\Phi^t(x_0)) = f(x_0) \quad \text{for all } t \text{ and a.e. } x_0 \in M$$

$$\Downarrow$$

$$f \text{ is constant a.e. on } M$$

- **E4:** The system is metrically indecomposable, namely for any measurable $A \subseteq M$
$$\forall t \;\; \Phi^t(A) = A \implies \mu(A) = 0 \text{ or } 1$$

Proof.

- **Proof of** $E2 \implies E1$**)** Let $A \subseteq M$ be a measurable set. Its indicator function satisfies $\chi_A \in L^1(M, \mu)$. Applying E2 to $f = \chi_A$ directly yields:

$$\langle \chi_A \rangle_t(x_0) = \langle \chi_A \rangle_p = \int_M \chi_A d\mu = \mu(A) \quad \text{for a.e. } x_0 \in M$$

  This is exactly E1.

- **Proof of** $E1 \implies E4$**)** Let A be an invariant set ($\forall t, \Phi^t(A) = A$). If $\mu(A) > 0$, there is some $x_0 \in A$ where the time average converges. Since the trajectory stays entirely in A, $\chi_A(\Phi^t(x_0)) = 1$ for all t. Evaluating the limit via E1 gives:

$$\mu(A) = \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} 1 = 1$$

  Thus, we must have either $\mu(A) = 0$ or $\mu(A) = 1$.

- **Proof of** $E4 \implies E2$**)** Assume E2 is false. By Birkhoff's Ergodic Theorem, the time average $\bar{f}(x) = \langle f \rangle_t(x)$ exists a.e., is invariant, and $\int_M \bar{f} d\mu = \langle f \rangle_p$. If E2 fails, the invariant set $A = \{x \in M \mid \bar{f}(x) > \langle f \rangle_p\}$ has $\mu(A) > 0$. By E4, $\mu(A) = 1$. This implies:

$$\int_M \bar{f} d\mu > \langle f \rangle_p$$

  This contradicts Birkhoff's theorem ($\int_M \bar{f} d\mu = \langle f \rangle_p$). Therefore, E2 must hold.

- **Proof of** $E4 \iff E3$**)**
  - $E4 \implies E3$**:** Let $f \in L^1(M, \mu)$ be invariant. For any $c \in \mathbb{R}$, the sub-level set $A_c = \{x \in M \mid f(x) < c\}$ is invariant. By E4, $\mu(A_c) \in \{0, 1\}$. Since $\mu(A_c)$ monotonically increases from 0 to 1, it must jump at exactly one value c_0. Therefore, $f(x) = c_0$ almost everywhere.
  - $E3 \implies E4$**)** Let $A \subseteq M$ be an invariant set. Its indicator function $\chi_A \in L^1(M, \mu)$ is invariant. By E3, χ_A is constant almost everywhere. Since it only takes values 0 or 1, we must have $\chi_A = 0$ a.e. ($\mu(A) = 0$) or $\chi_A = 1$ a.e. ($\mu(A) = 1$).

□

A dynamical system having this equivalent properties is called ergodic. Let us consider some examples.

Example: As we shown in Sec 3.1, mechanical systems satisfying the conditions of the Helmholtz theorem are ergodic, as they satisfy E2.

Example: An integrable Hamiltonian system is not ergodic as it possesses a lot of constants of motion from the very definition. Hence, as special cases, the Kepler problem for 2 bodies[2] and direct sum of harmonic oscillators are not ergodic.

Example: Let us investigate the ergodicity of the discrete translations on S^1 with $\alpha \in \mathbb{R}$. If $\alpha \in \mathbb{Q}$, then the system is periodic, hence do not possess E1 and is not ergodic. However, if $\alpha \notin \mathbb{Q}$, the system is ergodic. We will show this by showing E3.

Proof. Let $f \in L^1(S^1)$ be a constant of motion, meaning $f(x + \alpha) = f(x)$ a.e. Expressing f as a Fourier series:

$$f(x) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k x}$$

[2] It is also known that 3 body problem is not ergodic, despite being chaotic.

Applying the invariance $f(x+\alpha) = f(x)$:

$$\sum_{k\in\mathbb{Z}} c_k e^{2\pi i k(x+\alpha)} = \sum_{k\in\mathbb{Z}} c_k e^{2\pi i k x}$$

Equating the Fourier coefficients yields:

$$c_k e^{2\pi i k\alpha} = c_k \implies c_k(1 - e^{2\pi i k\alpha}) = 0$$

Since $\alpha \notin \mathbb{Q}$, we have $k\alpha \notin \mathbb{Z}$ for all $k \neq 0$, which means $e^{2\pi i k\alpha} \neq 1$. Therefore:

$$c_k = 0 \quad \text{for all } k \neq 0$$

Thus, $f(x) = c_0$, meaning f is constant a.e. This satisfies E3, establishing that the system is ergodic. □

Example: Consider the space $M = S^1 = \mathbb{R}/\mathbb{Z}$ with the Lebesgue measure μ. We define the doubling map $T : S^1 \to S^1$ as

$$T(x) = 2x \pmod 1$$

To prove this system is ergodic, we will use condition E3. Let $f \in L^1(S^1)$ be an invariant function, meaning $f(T(x)) = f(x)$ a.e. $x \in S^1$. Expressing f through its Fourier series, we obtain

$$\sum_{k\in\mathbb{Z}} c_k e^{4\pi i k x} = \sum_{k\in\mathbb{Z}} c_k e^{2\pi i k x}$$

By equating the Fourier coefficients on both sides, we deduce that $c_m = c_{m/2}$ if m is even, and $c_m = 0$ if m is odd. This establishes a recursive relation implying $c_k = c_{2k} = c_{4k} = \dots$ for any integer k.

Because $f \in L^1(S^1)$, we also know from the Riemann-Lebesgue lemma that we have

$$\lim_{|m|\to\infty} c_m = 0$$

Therefore, we must have $c_k = 0$ for all $k \neq 0$. This leaves only $f(x) = c_0$, meaning f is constant a.e. on S^1. This satisfies E3, establishing the system's ergodicity.

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